

SIEMENS



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WL Low Voltage Metal-Enclosed Switchgear

Selection and Application Guide

Answers for industry.

Type WL Low Voltage Metal-Enclosed Switchgear

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Type WL Low Voltage Metal-Enclosed Switchgear

General Information

Siemens Type WL low voltage metal-enclosed switchgear is designed, constructed and tested to provide superior power distribution, power monitoring and control. At the heart of the Type WL low voltage switchgear is the World Class Siemens WL breaker.

Siemens Type WL low voltage switchgear can be utilized in the following applications:

- **Industrial**
 - Heavy assembly
 - Semiconductor
 - Petrochemical
 - Automotive
 - Biotech
 - Pharmaceutical
 - **Institutional**
 - Water treatment
 - Airports
 - Universities
 - Medical facilities
 - Correctional facilities
 - **Critical Power**
 - Data Processing
 - Continuous industrial process
 - Hospitals
 - **Utility and co-generation**
 - **Commercial**
 - Large office buildings
 - Distribution centers
 - Large warehouses
- Product Scope:**
- **Equipment ratings**
 - 635VAC Maximum
 - 3 Phase 3 Wire,
 - 3 Phase 4 Wire
 - 50/60 Hz
 - 6000 amp maximum horizontal bus
 - 6000 amp maximum vertical bus
 - **Enclosure options**
 - NEMA 1 Indoor
 - NEMA 3R Outdoor Walk-In
 - NEMA 3R Outdoor Non Walk-in

Siemens WL breakers can be manually or electrically operated, fused or unfused and are available in the following rating designations – N, S, H, L, M and F. Refer to tables on Page 13 for interrupt and withstand ratings for each rating designation.



Type WL Low Voltage Metal-Enclosed Switchgear

Industry Standards

Type WL switchgear with power circuit breakers are designed, tested and constructed in accordance with:

- UL 1558 — Metal-Enclosed Low Voltage Power Circuit Breaker Switchgear
- ANSI C37.20.1 — Metal-Enclosed Low Voltage Power Circuit Breaker Switchgear
- ANSI C37.50 — Test Procedure for Low Voltage AC Power Circuit Breakers Used in Enclosures
- ANSI C37.51 — Conformance Testing of Metal-Enclosed Low Voltage AC Power Circuit Breaker Switchgear Assemblies
- NEMA SG5 - Power Switchgear Assemblies
- Applicable requirements of the National Electric Code (NEC)

WL drawout circuit breakers are in accordance with:

- UL 1066 — Low Voltage AC and DC Power Circuit Breakers Used in Enclosures
- ANSI C37.13 — Low Voltage AC Power Circuit Breakers Used in Enclosures
- ANSI C37.16 — Preferred Ratings, Related Requirements, and Application for Low Voltage Power Circuit Breakers and AC Power Circuit Protectors

- ANSI C37.17 — Trip Devices for AC and General Purpose DC Low-Voltage Power Circuit Breakers
- NEMA SG3 - Low Voltage Power Circuit Breakers

Features and modifications required by NEC are incorporated when the assembly is designated as "Service Equipment."

UL Listing

Underwriters' Laboratories listing mark (UL) is supplied for each vertical section provided all devices within a vertical section are UL Listed or UL Recognized and suitable for the intended use. All circuit breaker drawout elements are UL Listed.

Optional CSA compliance with cUL labeling is available.

Arc Resistant

Optional Type WL arc resistant low voltage switchgear is available and is UL listed to ANSI/IEEE C37.20.7. Type 2B arc resistant accessibility rating with maximum internal arcing short-circuit current rating of 100kA @508V and 85kA @ 635V.

Seismic Qualification

Seismic qualification to all major seismic construction standards (IBC, UBC, CBC, SBC, BOCA and IEEE 693) is available.

Type WL Low Voltage Metal-Enclosed Switchgear

Construction Details

General

The Siemens Type WL switchgear assembly consists of one or more metal-enclosed vertical sections. The end sections are designed to allow installation of future sections.

Each vertical section consists of up to four individually enclosed breaker or auxiliary compartments which are sized to provide uniform height.

Included in each assembly are various components such as circuit breakers, instrumentation and control equipment, transformers, relays, three-phase bus work, and all internal wiring, connectors, and other supporting equipment.

In accordance with ANSI C37.20.1, the maximum temperature for parts that are handled is 50°C. The main bus maximum temperature rise is 65°C above 40°C ambient. The temperature rise of the air surrounding the cable connection points is limited to 45°C above 40°C ambient.

Finish

During construction, the structural steel parts, panels, and compartments are all prepared for painting by a five-stage wash system.

Standard finish color is light gray ANSI 61. The standard painting process is a UL approved electrostatic powder coat paint system utilizing a polyester powder coat paint. The completed finish has a nominal 2 mils dry film thickness.

Assembly Construction

Siemens Type WL metal-enclosed low voltage switchgear is constructed of a rigid internal frame structure that minimizes the possibility of damage during shipment and supports multiple installation methods – rolling or lifting. Lifting eyes are integrated into the internal frame design and ensure the structural integrity of the lifting assembly is always adequate for the weight of the total structure.

If requested in advance, the switchgear structure can be shipped so that the unit can be tilted onto its back during installation. This is an option that must be specified at order entry.

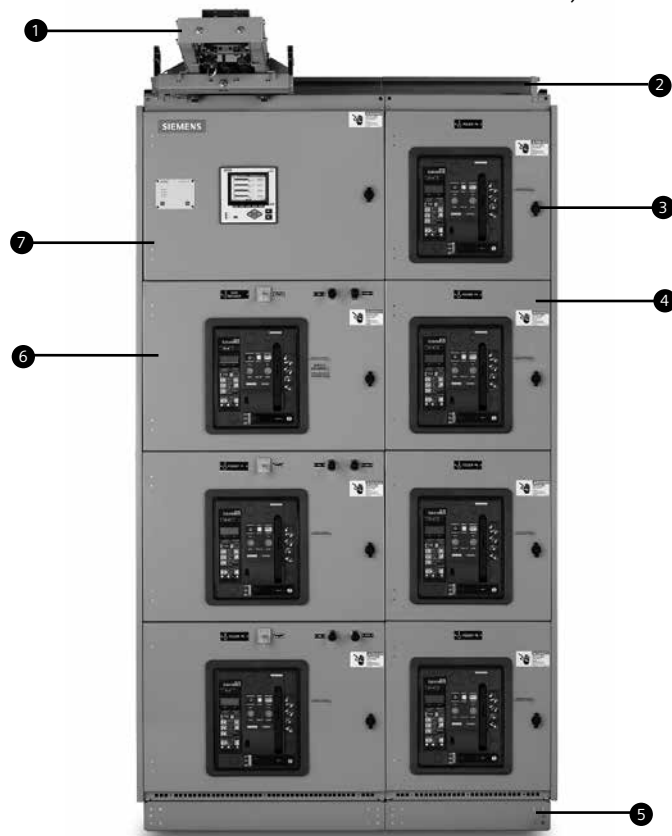
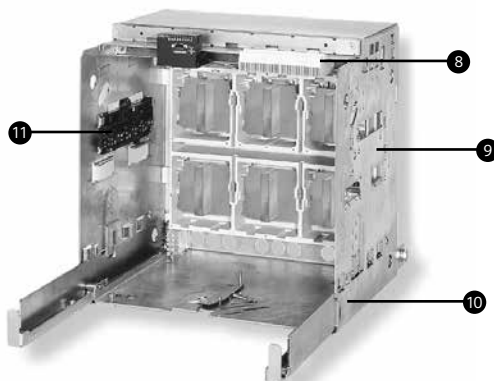
Each complete vertical section contains three compartments.

- (1) Front compartment containing breakers and/or auxiliary equipment
- (2) Bus compartment containing horizontal and vertical bus
- (3) Rear cable compartment containing the load side runbacks connecting the load side of the breaker to the load cable terminals

Within the front compartment, each breaker is barriered and compartmented from all other breakers in the front compartment. This design also isolates the breakers in the front compartment from the bus compartment.

Optional barriers can be supplied to isolate the bus compartment from the rear cable compartment. Other optional barriers include: (1) Full depth section barriers to isolate one section from the adjacent section(s). (2) Barriers to isolate the incoming line side connections to the main breaker(s) from the load side bus and connections in the switchgear section. (Line/load barriers are provided as a standard feature for service equipment main breakers.)

- 1 Breaker Hoist and Track
- 2 Ventilation and Lifting Structure
- 3 Quarter Turn Door Latch
- 4 Secondary Disconnect Access Door
- 5 Channel Sill Base (Optional)
- 6 Breaker Compartment
- 7 Auxiliary Instrument Compartment
- 8 Secondary Disconnect
- 9 Breaker Cradle (Guide Frame)
- 10 Breaker Drawout Rail
- 11 TOC Switch Operator



Type WL Low Voltage Metal-Enclosed Switchgear

Construction Details

Main and Ground Bus

The standard main bus is silver-plated copper. Tin-plated copper bus is optionally available. Vertical and horizontal bus bar utilize a channel shape design to maximize short circuit withstand capability and minimize heat rise. All bus joints include Grade 5 bolts and conical spring washers. Provisions for future extension of the main bus include plated joints and high tensile strength steel hardware.

The main three-phase horizontal bus is arranged vertically one phase above the other with edge-to-edge alignment to provide high, short circuit strength. Insulated main bus with isolated vertical bus is optional.

Vertical bus ratings available are 1600, 2000, 3200, 4000, 5000 and 6000 amperes continuous current. Horizontal bus ratings available are 1600, 2000, 3200, 4000, 5000 and 6000 amperes. A neutral bus is furnished when specified, and can be rated 1600, 2000, 3200, 4000, 5000 or 6000 amperes continuous current.

A 1/4" X 3" standard copper ground bus extends through all sections. Cable lugs are mounted to the ground bus in each section.

Standard short-circuit withstand (4 cycle) and short-time withstand (60 cycle) bus bracing is 100,000 amperes. Higher short-circuit withstand bus bracings (150kA and 200kA) are available.

Load side runbacks for feeder circuits are copper construction, are insulated with sleeve tubing in the main bus area, and are supported by high-strength bus bracing.

Control and Communication Wiring

Standard control and communication wiring is #14 AWG extra-flexible, stranded copper type SIS. Control and communication wiring is installed and accessed from the front of the switchgear structure. Each breaker compartment has a dedicated horizontal and vertical wireway.

For devices not having screw-type terminals, pressure terminals are used.

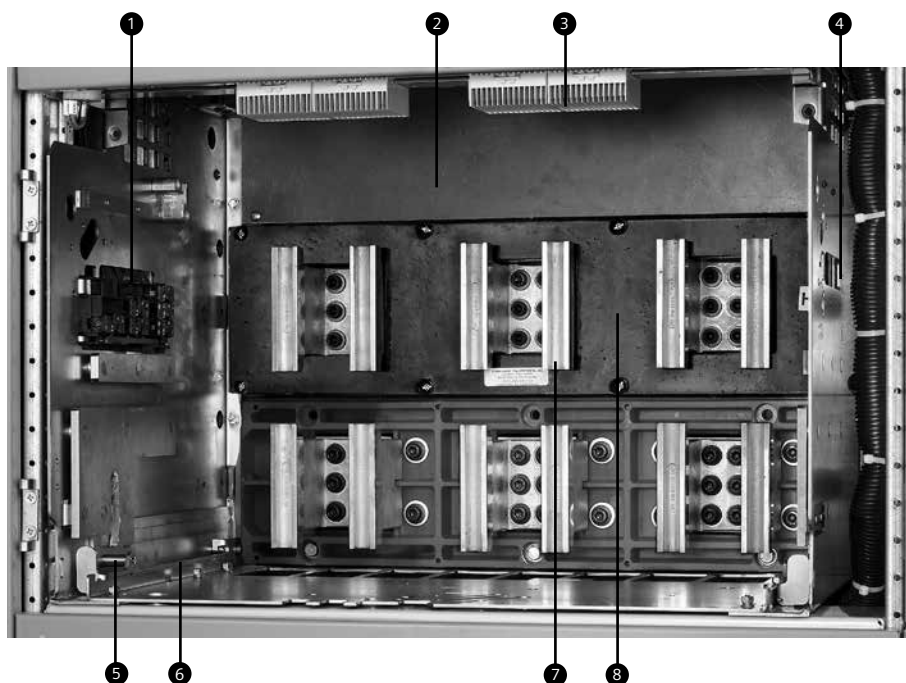
Insulation

The insulation used is a UL recognized thermoset material that has excellent heat resistance, flame retardance, dimensional stability and low moisture absorption.

Circuit Breaker Compartments

Typical circuit breaker compartments include primary disconnects, drawout rails, secondary disconnects, vertical wireway, horizontal wireway and, if applicable, TOC switch operator, MOC switch operator and associated interlocks. Draw-out rails allow the breaker to be withdrawn from the compartment without additional extensions or adapters. Up to six (2 sets of three) current transformers for metering or relaying can be mounted in each compartment.

A variety of auxiliary devices such as breaker control switches, indicating lights and pushbuttons can be mounted on the breaker compartment door.



- | | |
|------------------------------------|--------------------------------------|
| ① TOC Switch Operator | ⑤ Interference Interlock |
| ② Breaker Compartment Rear Barrier | ⑥ Drawout Rails |
| ③ Secondary Disconnect | ⑦ Primary Disconnect |
| ④ Vertical Wireway | ⑧ Cradle Mounted Current Transformer |

Circuit Breaker Cell Interior

Type WL Low Voltage Metal-Enclosed Switchgear

Construction Details

Options

Switchgear Mounted Hoist

The integrally mounted hoist, standard on walk-in outdoor and optional on indoor switchgear enclosures, travels along rails on top of the switchgear to assist in breaker handling.

TOC and MOC Switches

The Truck Operated Cell (TOC) Switch provides interlocking control or remote indication of the breaker racking position. The cubicle mounted auxiliary switch or Mechanism Operated Cell (MOC) switch provides interlocking control or remote indication based on the main contact position (open or closed).

Shutters

These provide protection against accidental contact with primary disconnects in a compartment when the breaker is removed. Shutters automatically close when the breaker is withdrawn and are pad-lockable and field installable.

Key Interlock

This provides a mechanical means for operating circuit breakers and other devices only when prescribed conditions are met.

Test Set

A portable breaker test set is available as an option and supports testing the full range of functions and protective settings supplied with the breaker trip unit.

Metering and Auxiliary Compartments

Compartments are available to house devices such as voltage transformers, metering, control power transformers, and supervisory devices.

Instrument and Control Transformers

Voltage transformers and control power transformers are mounted in auxiliary compartments. These transformers are protected by primary pull-out type current-limiting fuses and secondary fuses. Current transformers are normally mounted on the compartment primary disconnect studs where they are readily accessible. See Tables on Page 31 for available ratings.

Miscellaneous

- Each switchgear lineup includes a breaker lifting device that is adjustable for use with Size II and Size III breakers.
- An optional portable breaker hoist is available if the integrated breaker hoist and track is not specified.
- A test cabinet is also available as an option. The test cabinet is wall mounted necessary equipment for testing electrically-operated breakers that have been removed from the breaker compartment. The test cabinet doesn't include or replace a breaker trip unit tester.
- A WL remote breaker racking device (RBRD) is available as an optional accessory that allows maintenance personnel to safely rack Siemens Type WL breakers into the Connect, Test and Disconnect positions from up to 30 feet away from the breaker. This allows the operator to be outside the arc flash hazard boundary and thereby providing additional personnel protection.
- 4" high formed steel channel sills are available for indoor switchgear enclosures.

Outdoor Switchgear

Type WL switchgear is available in two outdoor (NEMA 3R) enclosures. Walk-in and non walk-in versions are available to meet your particular application.

For protection from snow, rain and other foreign matter, both outdoor enclosures rest on a six-inch high, formed steel base which provides rigid support and a tight bottom seal. A heavy duty protective under-coating is applied to the underside of all outdoor enclosures to protect against moisture and corrosion. Shielded ventilation housings permit proper air circulation while excluding dirt and foreign matter.

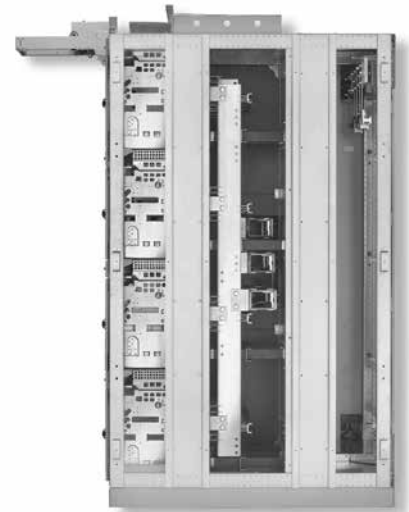
In the walk-in outdoor enclosure a lighted, unobstructed service aisle is provided at the front of the switchgear allowing inspection and maintenance without exposure to the elements. An access door equipped with an emergency bar release is located at each end of the aisle.

The following features are standard with walk-in outdoor enclosures.

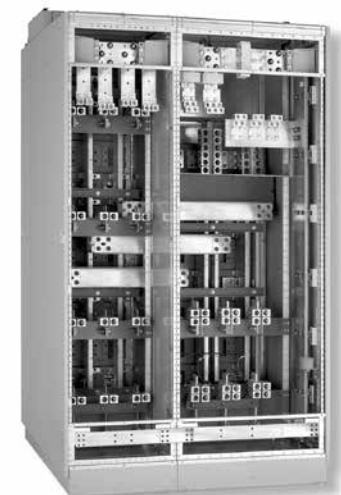
- (1) Space heaters in breaker compartment and bus compartment.

- (2) Screens and filters for exterior door ventilation louvers.
- (3) Incandescent lighting receptacle with three-way switch at each aisle access door.
- (4) Duplex receptacle with ground fault protection at each aisle access door.
- (5) Load center for power distribution to lights, receptacles, switches and heaters.

For non walk-in outdoor enclosures, space heaters and screens/filters for ventilation louvering are standard with lighting, receptacles, switches and load centers offered as options.



Side View



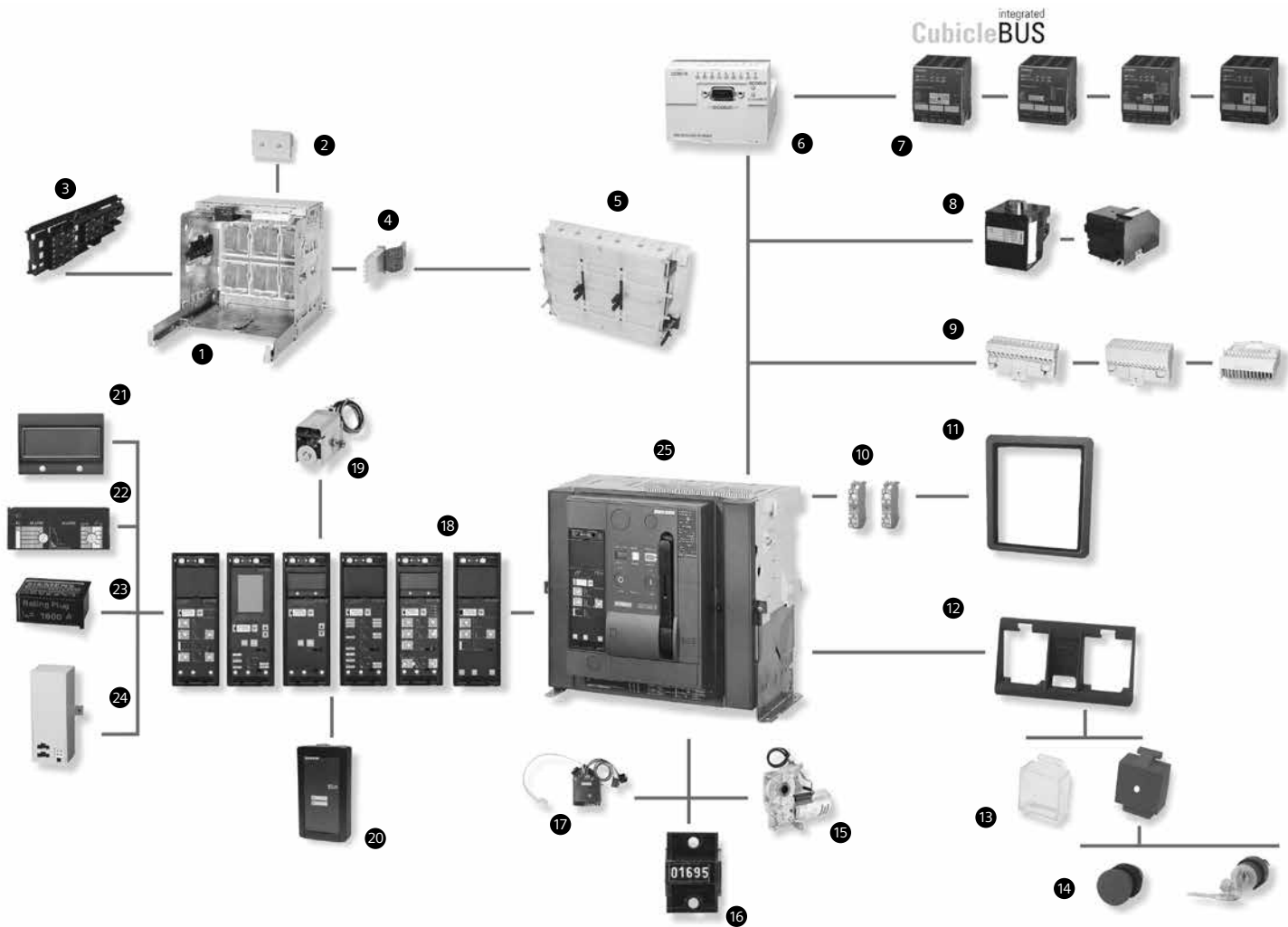
Rear View

Type WL Low Voltage Metal-Enclosed Switchgear

WL Circuit Breaker

WL Circuit Breaker:

Superior individual products for low-voltage power distribution systems



- 1 Guide Frame
- 2 Vertical to Horizontal BUS Connector
- 3 Position Signaling Switch (TOC)
- 4 Breaker / Guide Frame Grounding Contact
- 5 Shutter (locking)
- 6 MODBUS or PROFIBUS Communications
- 7 External CubicleBUS I/O Module
- 8 Plug-In Open and Closed Solenoids
- 9 Multiple Secondary Connections

- 10 Auxiliary Switch Block
- 11 Door Sealing Frame
- 12 Interlocking Set Base Plate
- 13 Protective Cover for OPEN/CLOSE Buttons
- 14 Multiple Key Locking Accessories
- 15 Single Bolt Motor Operator Installation
- 16 Operations Counter
- 17 Breaker Status Sensor (BSS)

- 18 Complete Trip Unit Family
- 19 Remote Reset
- 20 Breaker Data Adapter (BDA) for Internet Connection
- 21 Multi Angle LCD Module
- 22 Ground Fault Protection Module
- 23 Rating Plug
- 24 Metering Function (+ wave forms and harmonics)
- 25 Circuit Breaker

Type WL Low Voltage Metal-Enclosed Switchgear

Electronic Trip Unit

Electronic Trip Unit

During development of our electronic trip units we have consistently striven to ensure modularity. The following are just some of the modules that are simple to retrofit at any time:

- Ground fault protection
- Communication
- Metering function
- Displays
- Rating plugs

This enables fast local adaptation to new system conditions. At the same time, the ETUs are provided with new, innovative functions, and all trip units are completely interchangeable independent of breaker ratings.

Rating Plug

The Rating Plug is a replaceable module that enables users to reduce the rated device current for optimum adaptation to the system; e.g. during startup of a plant section. The Rating Plug should be selected so that it corresponds to the rated current of the system.

Switch-selectable I²t or I⁴t Characteristic Curve Improved Overload Protection

The best possible protection is assured when all protective devices in the system are optimally coordinated. To achieve optimum selectivity and coordination, the long-time characteristic can be switched between I²t and I⁴t.

Switchable Parameter Sets

To allow the protection to adapt to changes in system needs such as switching between utility and generator feeds, WL Circuit Breakers support ETUs with two independent parameter sets. Switching between the parameter sets occurs in less than 100 ms and can be done remotely or via a contact input to an optional CubicleBUS module.

Extended Instantaneous Protection

The electronic trip units designed for use with the WL circuit breaker provide a feature we call "Extended Instantaneous Protection" (Patent Pending).

It allows the WL breaker, as a family, across the entire range of ampacities to be applied at the withstand rating of the breaker with minus 0% tolerance; that means no instantaneous override. EIP further enables the circuit breaker to be applied up to the full interrupting rating of the breaker on systems where the available fault current exceeds the withstand rating, even with LS-only trip units. Why is this feature important? The answer is reliable power.

The coordination of the main breaker and the first level of feeder breakers is especially important because of the wide spread outage that will occur if one of these breakers trips unnecessarily.

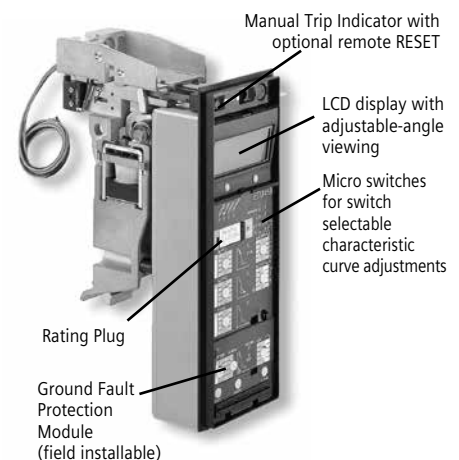
Conventional practice is to specify electronic trip breakers with "LS" type trip units in critical power systems. These 'Long-Time' and 'Short-Time' only trip units forgo the fast tripping times given by an 'Instantaneous' function. The justification for this delay is the benefit of allowing a downstream breaker to open first to clear a high magnitude fault. The main or feeder stays closed to keep the remainder of the loads operating.

However, a circuit breaker with an LS-only trip unit may never be applied on a system capable of delivering fault current higher than the breaker's withstand rating, commonly 85kA or less. Where the available fault current is above this level, a breaker with an additional function must be used — an instantaneous override. This instantaneous override function trips the breaker instantly when the fault current reaches a pre-determined level below the withstand rating, usually around 20% lower. The benefit of this override is to allow application of the breaker up to the interrupting rating, which may be as high as 150kA. The disadvantage is that it compromises the coordination benefit because the main will probably trip at the same time as a downstream branch breaker in that 20% lower override window.

This is where the Extended Instantaneous Protection feature of the WL can offer the next level of coordination and protection functionality. Unlike an instantaneous override, Extended Instantaneous Protection (EIP) allows the full withstand rating — in fact up to the tolerance of plus 20% higher. Of course, EIP still provides the ability of the breaker to be applied at the interrupting level, as high as 150kA in a Frame Size III, non-fused breaker. This unique combination enables the system designer to achieve the highest possible level of coordination in the industry and also allows application of the WL on modern power systems with extremely high levels of available fault current.

A further benefit offered by EIP, over a standard LS trip unit equipped breaker, is that it provides an extra measure of protection in the event that the available fault current increases at some time during the life of the system beyond the withstand level. This would typically be due to a utility transformer change but could also be due to the addition of generators or large motors that contribute fault current. EIP provides the breaker the ability to react in an instantaneous fashion to a high level fault instead of having to rely on the slower reaction time of the short-time function.

Sample Configuration of an ETU745



Type WL Low Voltage Metal-Enclosed Switchgear

Electronic Trip Unit



Selection Criteria for WL Circuit Breakers

The basic criteria for selecting circuit breakers is:

Maximum Available Short Circuit at the installation point. This value determines the short circuit current interrupting rating or short circuit current withstand rating of the circuit breaker.

Rated Current I_n which is to flow through the respective circuit breaker continuously. This value may not be greater than the maximum rated current of the circuit breaker. The rated current for the WL is determined by the rating plug, up to the maximum frame rating.

Ambient Temperature of the circuit breaker.

Design of the circuit breaker.

Protective Functions of the circuit breaker. These are determined by the selection of the appropriate trip unit.

Dynamic Arc-Flash Sentry (Patent Pending) A unique feature of the WL trip unit allows the system designer to achieve lower levels of arc flash energy and delayed tripping for selective trip coordination purposes.

Dynamic Arc-Flash Sentry (DAS) employs the unique dual protective setting capability of the 776 trip units, coupled with the ability to easily toggle to a lower arc flash parameter set. A normal operation parameter set can be optimized for selective trip coordination, while the second set is optimized for lower arc flash energy levels. The dynamic action comes from the ability to switch from the normal operation set to the arc flash limiting set based on the presence of personnel as they approach the flash protection boundary. A wide variety of switching methods may be used based on the needs of a particular facility. The capabilities range from fully automatic switching using appropriate occupancy sensors to manual switching via a key operation.

Type WL Low Voltage Metal-Enclosed Switchgear

Electronic Trip Unit



ETU745



ETU748



ETU776

Basic Protective Functions

Long-time overcurrent protection	L	●	●	●
Short-time delayed overcurrent protection	S	●	●	●
Instantaneous overcurrent protection	I	●	–	●
Neutral protection	N	●	●	●
Ground fault protection	G	○	○	○

Additional Functions

Selectable neutral protection	●	●	●
Defeatable short-time delay	●	●	●
Defeatable instantaneous protection	●	–	●
Selectable thermal memory	●	●	●
Zone selective interlocking	○	○	○
Selectable I ² t or fixed short-time delay	●	●	●
Adjustable instantaneous pick-up	●	–	●
Selectable I ² t or I ⁴ t long-time delay	●	●	●
Adjustable short-time delay and pick-up	●	●	●
Selectable and adjustable neutral protection	●	●	●
Dual protective setting capability	–	–	●
Extended instantaneous protection	●	●	●

Parameterization and Displays

Parameterization by rotary switches (10 steps)	●	●	–
Parameterization by communication (absolute values)	●	●	●
Parameterization by menu/keypad (absolute values)	–	–	●
Remote parameterization of the basic functions	–	–	●
Remote parameterization of the additional functions	–	–	●
Alphanumeric LCD	○	○	–
Graphical LCD	–	–	●

Metering Function

Metering function <i>Plus</i>	○	○	○
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Communication

CubicleBUS	●	●	●
Communication via PROFIBUS-DP	○	○	○
Communication via the MODBUS	○	○	○
Communication via the Ethernet (BDA)	○	○	○

● standard – not available ○ optional

Type WL Low Voltage Metal-Enclosed Switchgear

Time/Current Characteristic Curves

Tripping Characteristics

Every trip unit and every trip function has its own characteristic. You will find just a small section of these illustrated below. The characteristics show the respective greatest and smallest setting range of WL Circuit Breakers.

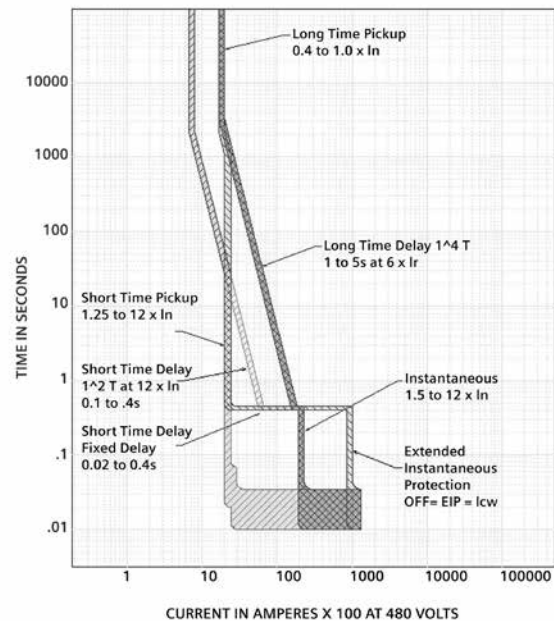
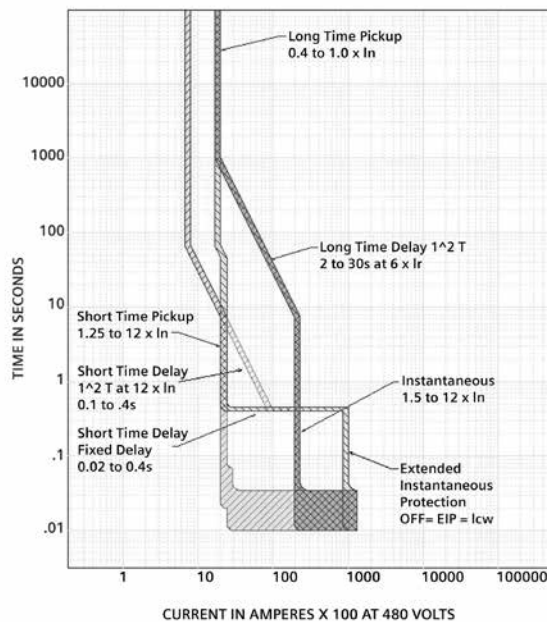
To obtain a complete release characteristic, the appropriate characteristic functions must be determined.

The characteristics show the behavior of the overcurrent release when it is activated

by a current already flowing before tripping. If the overcurrent trip takes place immediately after closing and the overcurrent release is therefore not yet activated, the opening time is prolonged by about 3 to 10 ms, depending on the value of the overcurrent.

Tripping Characteristics

ETU745

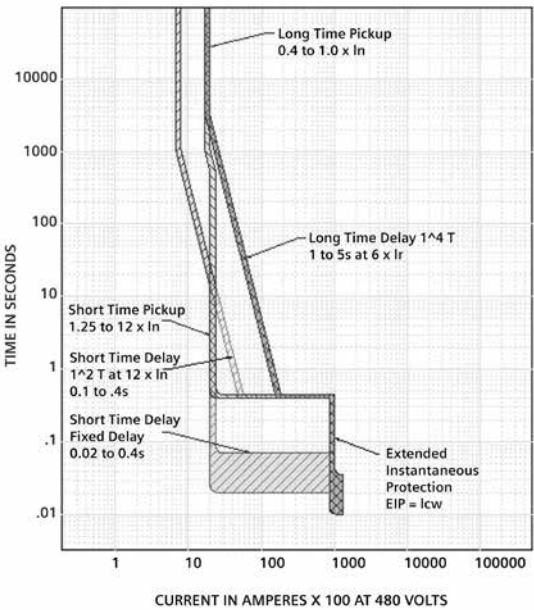
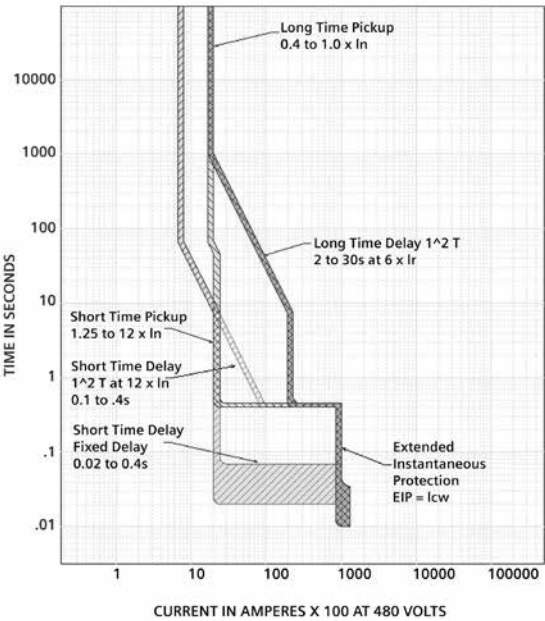


Type WL Low Voltage Metal-Enclosed Switchgear

Time/Current Characteristic Curves

Tripping Characteristics

ETU748

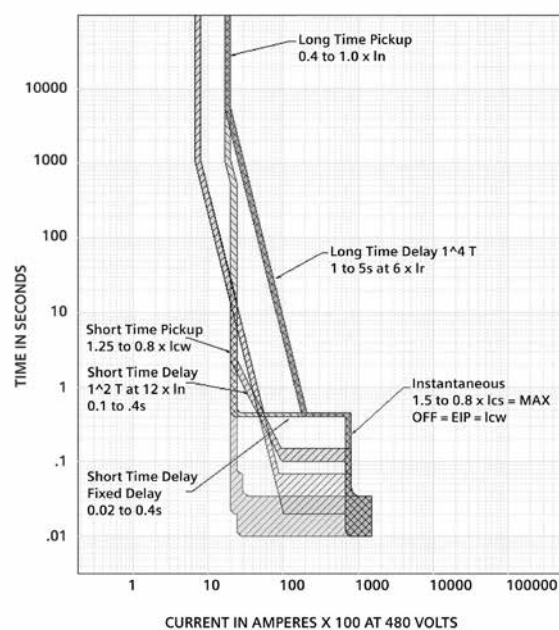
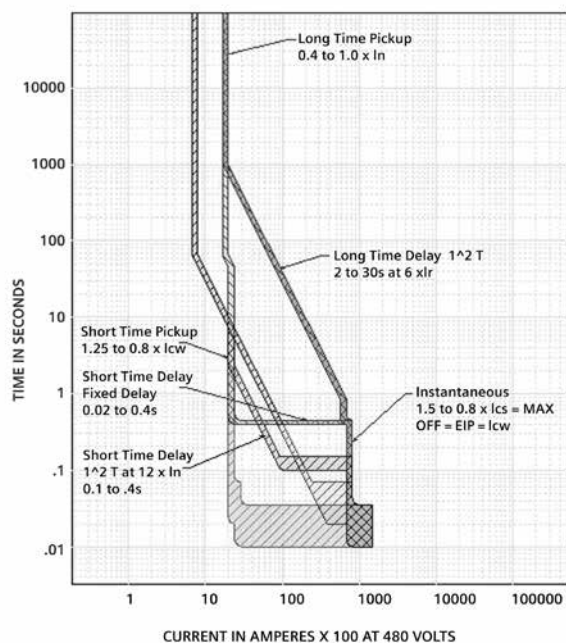


Type WL Low Voltage Metal-Enclosed Switchgear

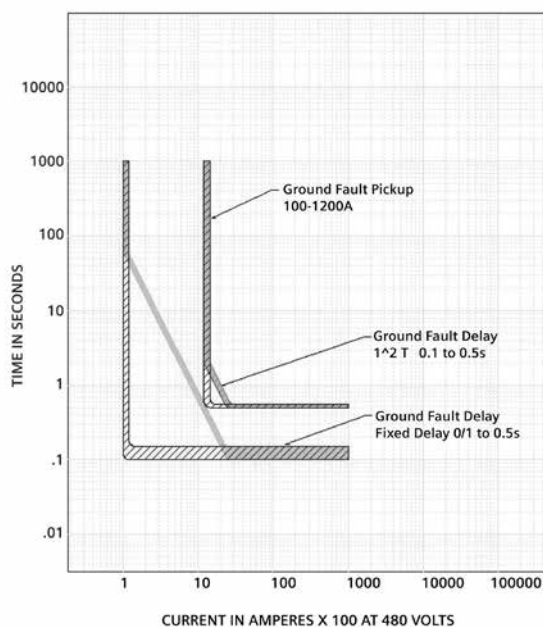
Time/Current Characteristic Curves

Tripping Characteristics

ETU776



Ground Fault Curve for ETU745, 748, and 776



Type WL Low Voltage Metal-Enclosed Switchgear

Breaker Technical Data

WL Circuit Breakers ANSI / UL 1066

Breaker Ratings

Frame Rating Rating Class		Frame Size II																	
		800					1600					2000				3200			
		N	S	H	L	F	N	S	H	L	F	S	H	L	F	S	H	L	
Instantaneous Short-circuit Current ¹ (kA RMS) 50/60 Hz	254VAC 508VAC 635VAC	50	65	85	100	200	50	65	85	100	200	65	85	100	200	65	85	100	
Short-time Withstand Current I _{cw} (kA RMS) 50/60 Hz	0.5s	50	65	65	85	—	50	65	65	85	—	65	65	85	—	65	65	85	
Extended Instantaneous Protection (kA RMS -0% to +20%)		50	65	65	85	—	50	65	65	85	—	65	65	85	—	65	65	85	
Close and Latch Ratings (kA RMS) 50/60 Hz		50	65	65	85	65	50	65	65	85	65	65	65	85	65	65	65	85	
Rating Plug Range		200, 225, 250, 300, 315, 350, 400, 450, 500, 600, 630, 700, 800 amps					200, 225, 250, 300, 315, 350, 400, 450, 500, 600, 630, 700, 800, 1000, 1200, 1250, 1600 amps					200, 225, 250, 300, 315, 350, 400, 450, 500, 600, 630, 700, 800, 1000, 1200, 1250, 1600, 2000 amps				200, 225, 250, 300, 315, 350, 400, 450, 500, 600, 630, 700, 800, 1000, 1200, 1250, 1600, 2000, 2500, 3000, 3200 amps			
Endurance Rating (switching operations with maintenance) ²	Mechanical Electrical	15,000 15,000					15,000 15,000					15,000 15,000				15,000 15,000			

Frame Rating Rating Class		Frame Size III												
		3200		4000				5000				6000		
		M	F	H	L	M	F	H	L	M	F	H	L	M
Instantaneous Short-circuit Current ¹ (kA RMS) 50/60 Hz	254V AC	150	200	85	100	150	200	85	100	150	200	85	100	150
	508V AC	150	200	85	100	150	200	85	100	150	200	85	100	150
	635V AC	85	200	85	85	85	200	85	85	85	200	85	85	85
Short-time Withstand Current I _{CW} (kA RMS) 50/60 Hz														
	0.5 s	100 ²	–	85	100 ³	100 ²	–	85	100	100 ³	–	85	100	100 ³
Extended Instantaneous Protection (kA RMS -0% to +20%)	254V AC													
	508V AC	150				150				150				150
	635V AC	85	–	85	100	85	–	85	100	85	–	85	100	85
Close and Latch Ratings (kA RMS) 50/60 Hz		100 ²	85	85	85	100 ²	85	85	85	100 ²	85	85	85	100 ²
Rating Plug Range		800, 1000, 1200, 1250, 1600, 2000, 2500, 3000, 3200 amps			800, 1000, 1200, 1250, 1600, 2000, 2500, 3000, 3200, 4000 amps				800,1000, 1200, 1250, 1600, 2000 2500, 3000, 3200, 4000, 5000 amps				800,1000, 1200, 1250, 1600, 2000 2500, 3000, 3200, 4000, 5000, 6000 amps	
Endurance Rating (switching operations with maintenance) ²	Mechanical Electrical	10,000 10,000			10,000 10,000				10,000 10,000				10,000 10,000	

¹ Maximum rated voltage for fused breakers is 600VAC.

² Maintenance means: replacing main contacts and arc chutes (see operating instructions).

³ Short-time withstand rating is 85kA RMS at 635VAC.

Type WL Low Voltage Metal-Enclosed Switchgear

Breaker Technical Data

WL Non-Automatic Switches ANSI / UL 1066

Ratings

Frame Rating	Rating Class	Frame Size II								Frame Size III					
		800		1600		2000		3200	3200	4000		5000		6000	
		L	F ¹	L	F ¹	L	F ¹	L	F ¹	L	F ¹	L	F ¹	L	
Short-time Withstand Current (kA RMS) 50/60 Hz	0.5 s	85	20	85	20	85	20	85	40	100	40	100	40	100	
Breaking Capacity with External Relay (kA RMS) 635Vac , 50/60 Hz, max time delay	0.5 s	85	20	85	20	85	20	85	40	100	40	100	40	100	

WL Circuit Breakers

Frame Rating		Frame Size II				Frame Size III			
		800	1600	2000	3200	3200	4000	5000	6000
Rated current I _n at 40°C, at 50/60Hz	A	800	1600	2000	3200	3200	4000	5000	6000
Rated operational (nominal) voltage	VAC	600	600	600	600	600	600	600	600
Rated maximum voltage	VAC	635	635	635	635	635	635	635	635
Permissible ambient temperature operation (for operation with LCD max. 55°C)	°C	-25 / +70	-25 / +70	-25 / +70	-25 / +70	-25 / +70	-25 / +70	-25 / +70	-25 / +70
Storage (observe special conditions for LCD)	°C	-40 / +70	-40 / +70	-40 / +70	-40 / +70	-40 / +70	-40 / +70	-40 / +70	-40 / +70
Power loss at Rated Current with 3-phase ² symmetrical load	W	85 130 (fused)	320 520 (fused)	500 850 (fused)	1150	700	1100	1650	2475
Operating times									
Make-time	ms	35	35	35	35	35	35	35	35
Break-time (with active ETU) ³	ms	34	34	34	34	34	34	34	34
Break-time (without active ETU) ⁴	ms	50	50	50	50	50	50	50	50
Total clearing time (with active ETU) ³	ms	50	50	50	50	50	50	50	50
Total clearing time (without active ETU) ⁴	ms	65	65	65	65	65	65	65	65
Make-time, electrical (via closing solenoid)	ms	50	50	50	50	50	50	50	50
Break-time, electrical (via shunt trip)	ms	40	40	40	40	40	40	40	40
(via instantaneous UVR)	ms	73	73	73	73	73	73	73	73

¹ Interrupting rating is equal to 200kA based on the rating of the fuse.

² Consult factory for fuse carriage power loss.

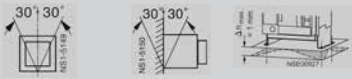
³ ETU with external control power.

⁴ ETU without external control power.

Type WL Low Voltage Metal-Enclosed Switchgear

Breaker Technical Data

WL Circuit Breakers

Frame Rating		Frame Size II				Frame Size III			
		800	1600	2000	3200	3200	4000	5000	6000
Endurance									
Mechanical (without maintenance)	operating cycles	12,500	12,500	10,000	10,000	5,000	5,000	5,000	5,000
Mechanical (with maintenance) ¹	operating cycles	15,000	15,000	15,000	15,000	10,000	10,000	10,000	10,000
Electrical (without maintenance)	operating cycles	7,500	7,500	4,000	4,000	2,000	2,000	2,000	2,000
Electrical (with maintenance) ¹	operating cycles	15,000	15,000	15,000	15,000	10,000	10,000	10,000	10,000
Switching frequency	1/h	60	60	60	60	60	60	60	60
Minimum interval between breaker trip and next closing of the circuit breaker (when used with the automatic mechanical reset of the bell alarm)	ms	80	80	80	80	80	80	80	80
Mounting position									
Auxiliary secondary wire size (Cu) max # of aux. connecting leads x cross section (solid or stranded)	Bare wire pressure terminal	1 x AWG 14 or 2 x AWG 16	1 x AWG 14 or 2 x AWG 16	1 x AWG 14 or 2 x AWG 16	1 x AWG 14 or 2 x AWG 16	1 x AWG 14 or 2 x AWG 16	1 x AWG 14 or 2 x AWG 16	1 x AWG 14 or 2 x AWG 16	1 x AWG 14 or 2 x AWG 16
	Tension spring terminal	2 x AWG 14	2 x AWG 14	2 x AWG 14	2 x AWG 14	2 x AWG 14	2 x AWG 14	2 x AWG 14	2 x AWG 14
	Ring tongue terminal (standard)	2 x AWG 14 1 x AWG 10 ² 2 x AWG 16	2 x AWG 14 1 x AWG 10 ² 2 x AWG 16	2 x AWG 14 1 x AWG 10 ² 2 x AWG 16	2 x AWG 14 1 x AWG 10 ² 2 x AWG 16	2 x AWG 14 1 x AWG 10 ² 2 x AWG 16	2 x AWG 14 1 x AWG 10 ² 2 x AWG 16	2 x AWG 14 1 x AWG 10 ² 2 x AWG 16	2 x AWG 14 1 x AWG 10 ² 2 x AWG 16
TOC wire connection size (Cu) max # of aux. connecting leads x cross section (solid or stranded)	Bare wire pressure terminal	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14
Weight ³									
Circuit Breaker	kg/lb	72/159	72/159	75/165	95/209	155/341	155/341	155/341	155/341
Guide Frame	kg/lb	51/112	51/112	60/132	69/152	139/306	139/306	139/306	139/306
MOC wire connection size (Cu) max # of aux. connecting leads x cross section (solid or stranded)	Bare wire pressure terminal	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14	1 x AWG 14

¹ Maintenance consists of replacing main contacts and arc chutes (see operating instructions.)

² For use only with Siemens supplied ring terminals (WL10RL).

³ Fused Breaker Weights

(kg/lb)	Frame Size II (fused)	Frame Size III (fused)
Breaker	103/227	same as table above
Guide Frame	68/150	130/275
Fuse Carriage	–	102/225

Type WL Low Voltage Metal-Enclosed Switchgear

Breaker Technical Data

WL Circuit Breaker Accessory Ratings

Manual Operating Mechanism with Mechanical Closing

Closing/charging stored energy mechanism

Maximum actuating force required on hand lever	52 lbs
Number of hand lever strokes required	9

Manual Operating Mechanism with Mechanical and Electrical Closing

Charging stored-energy mechanism

Closing solenoid and Shunt Trip	Coil voltage tolerance	24V DC	14 - 28V DC
		48V DC	28 - 56V DC
		120V AC / 125V DC	70 - 140V DC
			104 - 127V AC
		240V AC / 250V DC	140 - 280V DC
			08 - 254V AC
			180Y / 104V AC
			220Y / 127V AC
	Power consumption (5 % duty cycle)		120 W
	Minimum closing solenoid actuation signal required		50 ms

Motor Operating Mechanism with Mechanical and Electrical Closing

Spring charging motor

Motor voltage tolerance at 120V AC, 240V AC	85 - 110%
Extended tolerance for battery operation at 24V DC, 48V DC, 125V DC, 250V DC	70 - 126%
Power consumption of the motor	110 W
Time required for charging the stored-energy mechanism	≤ 10 s

Closing solenoid

For motor and closing solenoid
short-circuit protection

Short-circuit protection Standard slow-blow cartridge fuse	24 - 60 V	6A
	110 - 250 V	3A

Auxiliary Release

Undervoltage release (UVR)	Operating values	≥ 85% (circuit breaker can be closed)	
		35 - 70% (circuit breaker opens)	
	AC Coil voltage tolerance at 120V AC, 240V AC		85 - 110%
	DC Extended tolerance for battery operation at 24V DC, 48V DC, 125V DC, 250V DC		85 - 126%
Rated control supply voltage	AC 50/60Hz	V	120, 240
	DC	V	24, 48, 125, 250
Power consumption (inrush / continuous)	AC	VA	200 / 5
	DC	W	200 / 5
Opening time of the circuit breaker for AC / DC		ms	200
UVR (no time delay), 2 settings			
Setting 1		ms	80
Setting 2		ms	200
UVR (with time delay)			
Adjustable delay		s	0.2 to 3.2
Reset by additional NC direct opening		ms	≤ 100

Type WL Low Voltage Metal-Enclosed Switchgear

Breaker Technical Data

WL Circuit Breaker Accessory Ratings

Auxiliary Contacts and Mechanism Operated Contacts (MOC)				
Contact rating	Alternating current			
	50/60 Hz	Rated operational voltage	240V	
		Rated operational current, continuous	10A	
		Rated operational current, making	30A	
		Rated operational current, breaking	3A	
	Direct current	Rated operational voltage	24V, 125V, 250V	
		Rated operational current, continuous	5A	
		Rated operational current, making	1.1A at 24V, 1.1A at 125V, 0.55A at 250V	
		Rated operational current, breaking	1.1A at 24V, 1.1A at 125V, 0.55A at 250V	
Bell Alarm Switch and Ready-to-Close Signal Contact				
Contact rating	Alternating current			
	50/60 Hz	Rated operational voltage	240V	
		Rated operational current, continuous	5A	
		Rated operational current, making	8A	
		Rated operational current, breaking	5A	
	Direct current	Rated operational voltage	24V, 48V, 125V	250V DC
		Rated operational current, continuous	0.4A	0.2A
		Rated operational current, making	0.4A	0.2A
		Rated operational current, breaking	0.4A	0.2A
Shunt Trip, UVR and Blown Fuse Signaling Contacts				
Contact rating	Alternating current			
	50/60 Hz	Rated operational voltage	127V, 240V	
		Rated operational current, continuous	3A	
		Rated operational current, making	5A	
		Rated operational current, breaking	6A	
	Direct current	Rated operational voltage	24V, 48V, 125V	125V DC (IEC Rating Only)
		Rated operational current, making	1.0A	0.5A
		Rated operational current, breaking	1.0A	0.5A
	Position Signal Contact on the Guide Frame (TOC)			
Breaker position:	Connected position	3 form C	1 form C	6 form C
	Test position	2 form C	or 1 form C	or 0 form C
	Disconnected position	1 form C	1 form C	0 form C
Contact rating	Alternating current			
	50/60 Hz	Rated operational voltage	120V	240V
		Rated operational current, continuous	10A	10A
		Rated operational current, making	6A	3A
		Rated operational current, breaking	6A	3A
	Direct current	Rated operational voltage	24V	48V, 125V
		Rated operational current, continuous	6A	1A
		Rated operational current, making	6A	0.22A
		Rated operational current, breaking	6A	0.22A

Type WL Low Voltage Metal-Enclosed Switchgear

Breaker Technical Data

Function Overview of the Electronic Trip Units

Basic Functions		ETU745
	Long-time overcurrent protection Function can be switched ON/OFF Setting range $I_R = I_N \times \dots$	✓ — 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.8, 0.9, 1
	L Switch-selectable overload protection (I^2t or I^4t dependent function) Setting range of time delay class t_R at I^2t (seconds) Setting range of time delay t_R at I^4t (seconds) Thermal memory Phase loss sensitivity	✓ 2, 3.5, 5.5, 8, 10, 14, 17, 21, 25, 30 1, 2, 3, 4, 5 ✓ (via slide switch) at $t_{sd}=20ms$ (M)
	N Neutral protection Function can be switched ON/OFF N-conductor setting range $I_N = I_N \times \dots$	✓ ✓ (via slide switch) 0.5 ... 1
	S Short-time delayed overcurrent protection Function can be switched ON/OFF Setting range $I_{sd} = I_N \times \dots$ Setting range of time delay t_{sd} , fixed (seconds) Switch-selectable short-time delay short-circuit protection (I^2t dependent function) Setting range of time delay t_{sd} at I^2t (seconds) Zone Selective Interlocking (ZSI) function	✓ ✓ (via rotary switch) 1.25, 1.5, 2, 2.5, 3, 4, 6, 8, 10, 12 0.02 (M), 0.1, 0.2, 0.3, 0.4, OFF ✓ (via rotary switch) 0.1, 0.2, 0.3, 0.4 per CubicleBUS module
	I Instantaneous overcurrent protection Function can be switched on/off, Extended Instantaneous Protection is enabled when OFF Setting range $I_i = I_N \times \dots$	✓ ✓ (via rotary switch) 1.5, 2.2, 3, 4, 6, 8, 10, 12 $0.8 \times I_{cw} = \max$, OFF = $I_{cw} = EIP^1$
	G Ground fault protection ² Trip and alarm function Detection of the ground fault current by residual summing method Detection of the ground fault current by direct summing method Setting range of the I_g for trip Setting range of the I_g for alarm Setting range of the time delay (seconds) Switch-selectable ground fault protection (I^2t / fixed) Setting range time delay t_g at I^2t ZSI ground function	☆ (field installable module) ✓ ✓ ✓ A, B, C, D, E A, B, C, D, E 0.1, 0.2, 0.3, 0.4, 0.5 ✓ 0.1, 0.2, 0.3, 0.4, 0.5 per CubicleBUS module

¹ Extended Instantaneous Protection (EIP) allows the WL breaker to be applied at the withstand rating of the breaker with minus 0% tolerance; that means no instantaneous override whatsoever. EIP further enables the circuit breaker to be applied up to the full instantaneous rating of the breaker on systems where the available fault current exceeds the withstand rating.

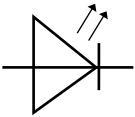
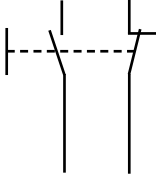
² Ground Fault Module cannot be removed after installation.

✓ available
— not available
○ optional

Type WL Low Voltage Metal-Enclosed Switchgear

Breaker Technical Data

Function Overview of the Electronic Trip Units

Basic Functions		ETU745
Parameter sets		
	Selectable between parameter set A and B	–
LCD		
	LCD, alphanumeric (4-line)	○
	LCD, graphic	–
Communication		
	CubicleBUS integrated	✓
	Communication capability via MODBUS or PROFIBUS	✓
Metering function		
	Metering function capability with Metering Function PLUS	✓
Display by LED		
	Trip unit active	✓
	Alarm	✓
	ETU error	✓
	L trip	✓
	S trip	✓
	I trip	✓
	N trip	✓
	G trip	✓ (only with ground fault module)
	G alarm	✓ (only with ground fault module)
	Tripped by extended protection or protective relay function	✓
	Communication	✓
Signal contacts with external CubicleBUS modules (Opto or relay)		
	Overcurrent warning	✓
	Load shedding OFF/ON	✓
	Early signal of long time trip (200ms)	✓
	Temperature alarm	✓
	Phase unbalance	✓
	Instantaneous trip	✓
	Short-time trip	✓
	Long-time trip	✓
	Neutral conductor trip	✓
	Ground fault protection trip	✓ (only with ground fault module)
	Ground fault alarm	✓ (only with ground fault module)
	Auxiliary relay	✓
	ETU error	✓

Step for Settings via Communications or ETU Key Pad

from ... to	step	from ... to	step
0 ... 1	0.1	1000 ... 1600	50
1 ... 100	1	1600 ... 10000	100
100 ... 500	5	10000 ... max	1000
500 ... 1000	10		

Setting range of the I_g

	Frame Size II	Frame Size III
A	100 A	400 A
B	300 A	600 A
C	600 A	800 A
D	900 A	1000 A
E	1200 A	1200 A

- ✓ available
- not available
- optional

Breaker Technical Data

Basic Functions		ETU748	ETU776
	L Long-time overcurrent protection Function can be switched ON/OFF Setting range $I_R = I_N \times \dots$ Switch-selectable overload protection (I^2t or I^4t dependent function) Setting range of time delay class t_R at I^2t (seconds) Setting range of time delay t_R at I^4t (seconds) Thermal memory Phase loss sensitivity	✓ – 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.8, 0.9, 1 ✓ 2, 3.5, 5.5, 8, 10, 14, 17, 21, 25, 30 1, 2, 3, 4, 5 ✓ (via slide switch) at $t_{SD}=20\text{ms}$ (M)	✓ – 0.4 ... 1 (step: 1A) ✓ 2 ... 30 (step: 0.1s) 1 ... 5 (step: 0.1s) ✓ (on/off via key pad or communications) ✓ (on/off via key pad or communications)
	N Neutral protection Function can be switched ON/OFF N-conductor setting range $I_N = I_N \times \dots$	– – –	✓ ✓ (via slide switch) 0.5 ... 1
	S Short-time delayed overcurrent protection Function can be switched ON/OFF Setting range $I_{SD} = I_N \times \dots$ Setting range of time delay t_{SD} , fixed (seconds) Switch-selectable short-time delay short-circuit protection (I^2t dependent function) Setting range of time delay t_{SD} at I^2t (seconds) Zone Selective Interlocking (ZSI) function	✓ ✓ (via rotary switch) 1.25, 1.5, 2, 2.5, 3, 4, 6, 8, 10, 12 M, 0.1, 0.2, 0.3, 0.4 ✓ (via rotary switch) 0.1, 0.2, 0.3, 0.4 per CubicleBUS module	✓ ✓ (via key pad or communications) 1.25 ... $0.8 \times I_{CW} = \text{max}$ (step: 10A) M, 0.08 ... 0.4, OFF (step: 0.001s) ✓ (via key pad or communications) 0.1 ... 0.4 (step: 0.001s) per CubicleBUS module
	I Instantaneous overcurrent protection Function can be switched ON/OFF, Extended Instantaneous Protection is enabled when OFF Setting range $I_I = I_N \times \dots$	– – $I_I = I_{CW} = \text{EIP } 1$	✓ ✓ (via key pad or communications) 1.5 $\times I_N$... $0.8 \times I_{CS} = \text{max}$, OFF = $I_{CW} = \text{EIP } 1$
	G Ground fault protection ² Trip and alarm function Detection of the ground fault current by residual summing method Detection of the ground fault current by direct summing method Setting range of the I_g for trip Setting range of the I_g for alarm Setting range of the time delay t_g (seconds) Switch-selectable ground fault protection (I^2t / fixed) Setting range time delay t_g at I^2t ZSI ground function	○ (field installable module) ✓ ✓ ✓ A, B, C, D, E A, B, C, D, E 0.1, 0.2, 0.3, 0.4, 0.5 ✓ 0.1, 0.2, 0.3, 0.4, 0.5 per CubicleBUS module	○ (field installable module) ✓ (via key pad or communications) ✓ ✓ A ... E (step: 1A) A ... E (step: 1A) 0.1 ... 0.5 (step: 0.001s) ✓ 0.1 ... 0.5 (step: 0.001s) per CubicleBUS module

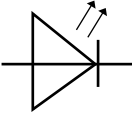
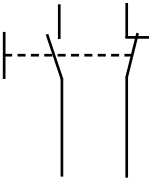
- ✓ available
- not available
- optional

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Type WL Low Voltage Metal-Enclosed Switchgear

Breaker Technical Data

Function Overview of the Electronic Trip Units

Basic Functions		ETU748	ETU776
Parameter sets			
	Selectable between parameter set A and B	–	✓
LCD			
	LCD, alphanumeric (4-line)	○	–
	LCD, graphic	–	✓
Communication			
	CubicleBUS integrated	✓	✓
	Communication capability via MODBUS or PROFIBUS	✓	✓
Metering function			
	Metering function capability with Metering Function PLUS	✓	✓
Display by LED			
	Trip unit active	✓	✓
	Alarm	✓	✓
	ETU error	✓	✓
	L trip	✓	✓
	S trip	✓	✓
	I trip	–	✓
	N trip	✓	✓
	G trip	✓ (only with ground fault module)	✓ (only with ground fault module)
	G alarm	✓ (only with ground fault module)	✓ (only with ground fault module)
	Tripped by extended protection or protective relay function	✓	✓
	Communication	✓	✓
Signal contacts with external CubicleBUS modules (Opto or relay)			
	Overcurrent warning	✓	✓
	Load shedding OFF/ON	✓	✓
	Early signal of long time trip (200ms)	✓	✓
	Temperature alarm	✓	✓
	Phase unbalance	✓	✓
	Instantaneous trip	✓	✓
	Short-time trip	✓	✓
	Long-time trip	✓	✓
	Neutral conductor trip	✓	✓
	Ground fault protection trip	✓ (only with ground fault module)	✓ (only with ground fault module)
	Ground fault alarm	✓ (only with ground fault module)	✓ (only with ground fault module)
	Auxiliary relay	✓	✓
	ETU error	✓	✓

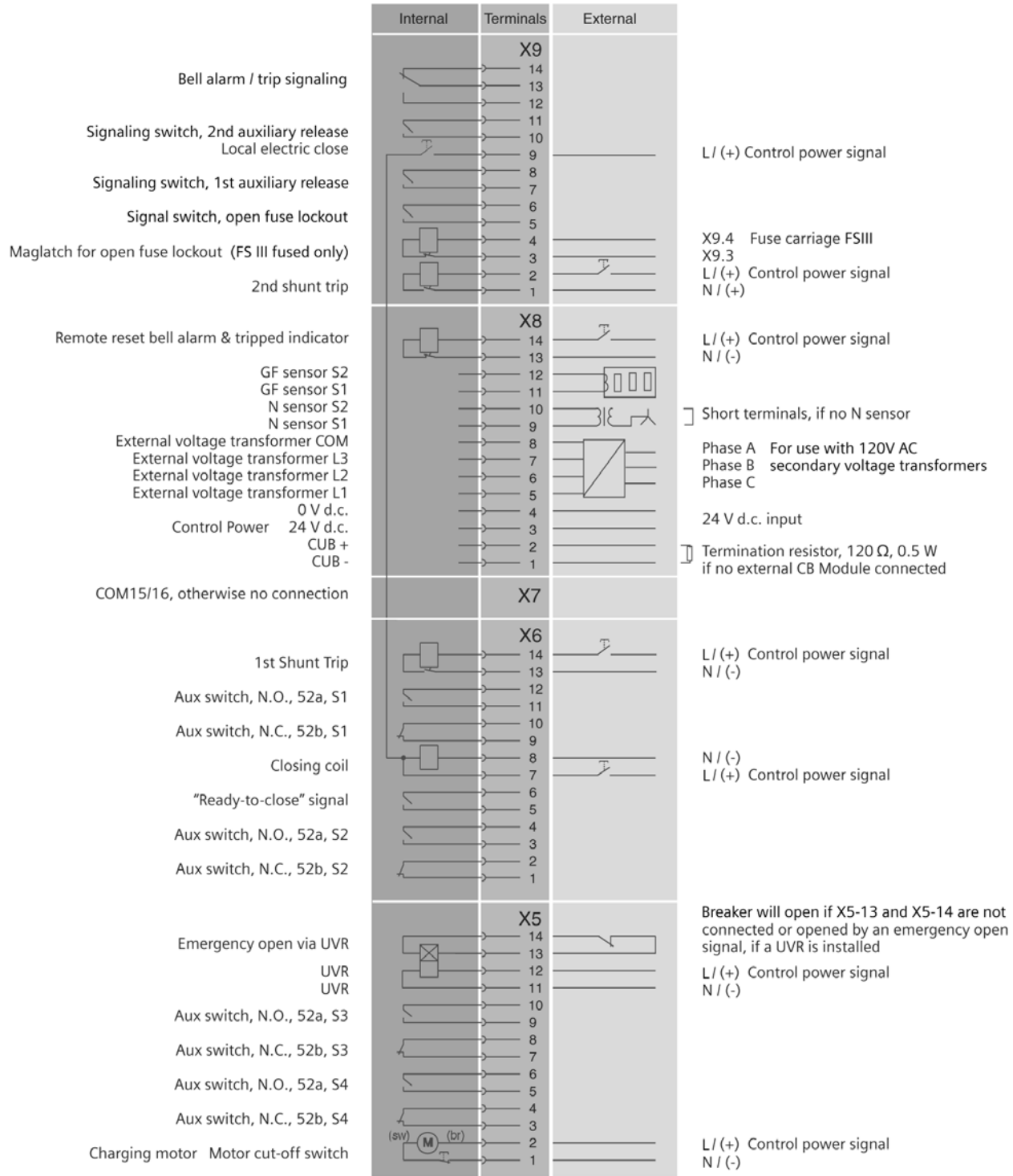
Metering and Protective Relaying Accuracies

Protective Relaying	Pick-up Accuracy
Phase Unbalance (I)	2% (5...50% I_N)
Phase Unbalance (V)	2% (5...50% V_N)
THD (I) (up to 29th)	+/- 3% (80...120% V_N)
THD (V) (up to 29th)	+/- 3% (80...120% V_N)
Overvoltage	+/- 2% (80...120% V_N)
Undervoltage	+/- 2% (80...120% V_N)
Under/Over Frequency	+/- 0.1 Hz

Metering Values	Accuracy
(I) at 1 x I_N	+/- 1%
(V) at 1 x V_N	+/- 0.5%
(P) at 1 x I_N	+/- 3%
(S) at 1 x I_N	+/- 2%
(Q) at 1 x I_N	+/- 3%

Type WL Low Voltage Metal-Enclosed Switchgear

WL Secondary Terminal Assignments

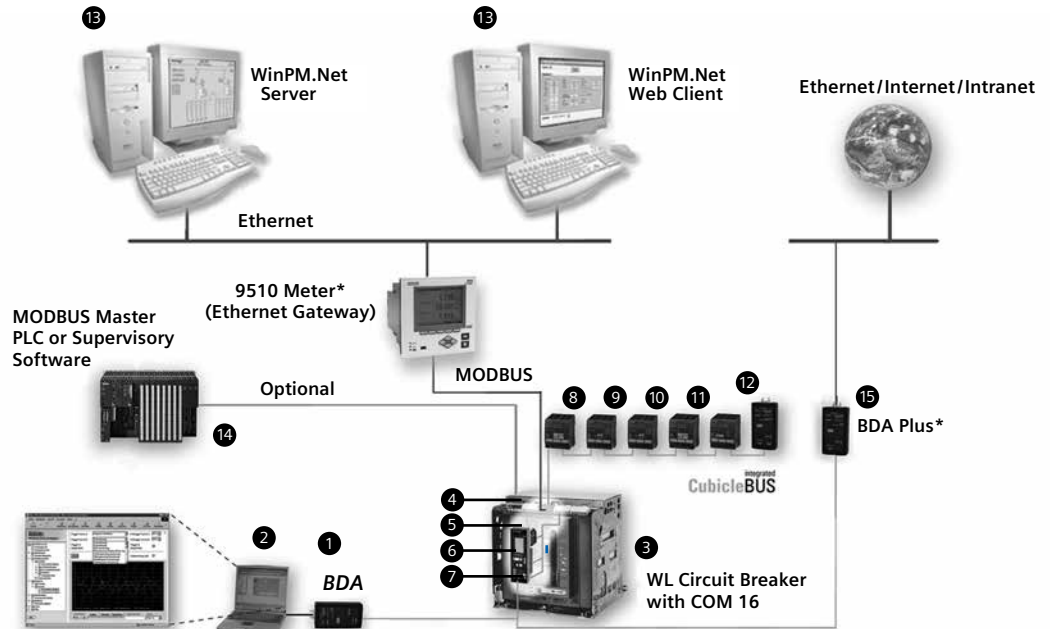


Type WL Low Voltage Metal-Enclosed Switchgear

WL Communication Overview

Connection Diagram

- 1 Breaker Data Adapter (BDA)
- 2 Browser-capable input and output device (e.g. notebook)
- 3 WL Circuit Breaker
- 4 COM 16 MODBUS module or COM 15 PROFIBUS module
- 5 Breaker Status Sensor (BSS)
- 6 Electronic Trip Unit
- 7 Metering function PLUS
- 8 Zone Selective Interlocking (ZSI) module
- 9 Digital output module with relay or optocoupler outputs
- 10 Digital output module with relay or optocoupler outputs, remotely configurable
- 11 Analog output module
- 12 Digital input module
- 13 WinPM.Net on PC
- 14 PLC (e.g. SIMATIC S7)
- 15 BDA Plus



* The Siemens BDA Plus or meters, 9330, 9350, 95/9610 can be used as a gateway to enable Ethernet communication to the WL Circuit Breaker.

Features

- Industry standard MODBUS or PROFIBUS communication available on all WL breakers from 200A to 6000A.
- The high modularity of the WL Circuit Breakers and accessories allows simple retrofitting of all communication components.
- The ability to connect additional input and output modules to the breaker-internal CubicleBUS of the WL opens up a range of opportunities to reduce secondary device count and wiring and to increase functionality implemented in switchgear.
- Innovative software products for local configuration, operation, monitoring and diagnostics of WL Circuit Breakers using MODBUS, PROFIBUS or via Ethernet/Intranet/Internet.
- Complete integration of WL Circuit Breakers in all Totally Integrated Power and Totally Integrated Automation Solutions.

Type WL Low Voltage Metal-Enclosed Switchgear

Section Configurations

General Notes:

- A blank/instrument compartment can always be substituted for a breaker compartment.
- Any 22" wide section can be 32" wide if more conduit working room is needed.
- For bus duct connections – if incoming is top, Compartment A must be blank/instrument, if incoming is bottom, Compartment D must be blank/instrument.
- Bussed transition section is 22" wide
- For close coupled transformer connections, Compartment A must be blank/instrument.
- Utility metering is always in a separate section. Section width is dependent on utility.

Switchgear Depth Dimensional Information

(Dimensions below are for internal frames – not total structure depth)

- Non-fused indoor – 60" standard, 70" and 80" optional
- Fused indoor – 65" standard, 75" and 80" optional
- Non-fused non-walk-in outdoor – 60" standard and 75" optional
- Fused non-walk-in outdoor – 65" standard and 75" optional
- Non-fused walk-in outdoor – 60" standard and 75" optional
- Fused walk-in outdoor – 65" standard and 75" optional
- Walk-in outdoor aisle is 42" deep
- Sections with cable connected main, tie and/or feeder breakers that are 3200 amp or greater must be minimum depth of 70" for unfused breakers and 75" for fused breakers.

Note 1 – If a 4000 amp feeder breaker is installed in Compartment C, Compartment D must be a Blank or Instrument Compartment.

Note 2 – If a 4000 amp breaker is installed in Compartment B, Compartment A must be a Blank or Instrument Compartment.

Note 3 – If incoming is bottom, feeder breakers can mount in compartments A and/or B.

Note 4 – If a 3200 amp breaker is installed in Compartment B, the middle level through bus is not available.

Note 5 – If a 3200 amp breaker is installed in Compartment D, the lower level through bus is not available.

Note 6 – Only one 800, 1600, 2000 amp feeder breaker can be mounted per section. If the horizontal main bus is at the top of the section, the 800, 1600, 2000 amp feeder breaker can go in the A compartment and a blank/instrument compartment must go in the D compartment. If the horizontal main bus is at the bottom of the section, the 800, 1600, 2000 amp feeder breaker can go in the D compartment and a blank/instrument compartment must go in the A compartment.

Note 7 – Any feeder section (or bus transition section) with 6000 amp vertical bus must be 32" wide.

Section Compartment Arrangement

A Compartment
B Compartment
C Compartment
D Compartment

Main Sections – Non-Fused Breakers

Blank or Instrument Compartment	Feeder Breaker 800, 1600, 2000	Blank or Instrument Compartment	Blank or Instrument Compartment	Blank or Instrument Compartment	Auxiliary Compartment
Main Breaker 800, 1600, 2000, 3200	Feeder Breaker 800, 1600, 2000, 3200	Blank or Instrument Compartment	Main Breaker 4000, 5000	Blank or Instrument Compartment	Main Breaker 6000
Feeder Breaker 800, 1600, 2000, 3200	Main Breaker 800, 1600, 2000, 3200	Main Breaker 800, 1600, 2000, 3200	Feeder Breaker 800, 1600, 2000, 3200, 4000 Refer to Note 1	Main Breaker 4000, 5000 Refer to Note 3	
Feeder Breaker 800, 1600, 2000	Blank or Instrument Compartment	Feeder Breaker 800, 1600, 2000	Feeder Breaker 800, 1600, 2000 Refer to Note 1	Feeder Breaker 800, 1600, 2000	Auxiliary Compartment
22 in. (559 mm)	22 in. (559 mm)	22 in. (559 mm)	32 in. (813 mm)	32 in. (813 mm)	32 in. (813 mm)

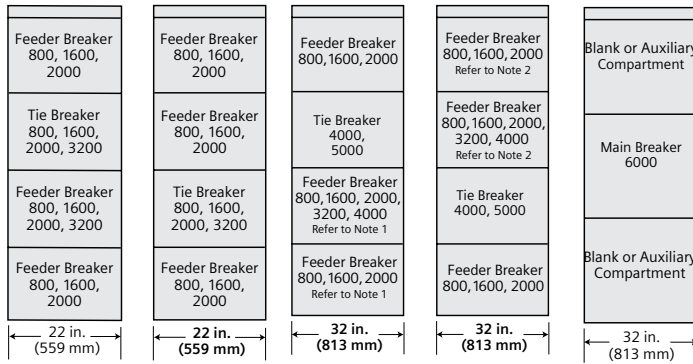
Main Sections – Fused Breakers

Blank or Instrument Compartment	Blank or Instrument Compartment	Blank or Instrument Compartment	Blank or Instrument Compartment
Main Breaker 800, 1600, 2000	Blank or Instrument Compartment	Main Breaker 3200, 4000	Main Breaker 5000
Feeder Breaker 800, 1600, 2000	Main Breaker 800, 1600, 2000		
Feeder Breaker 800, 1600, 2000	Feeder Breaker 800, 1600, 2000	Feeder Breaker 800, 1600, 2000	
22 in. (559 mm)	22 in. (559 mm)	32 in. (813 mm)	32 in. (813 mm)

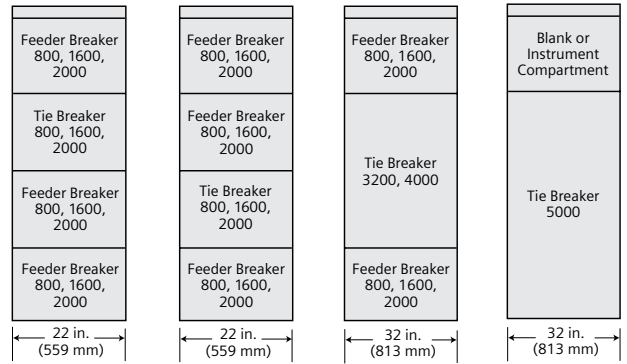
Type WL Low Voltage Metal-Enclosed Switchgear

Section Configurations

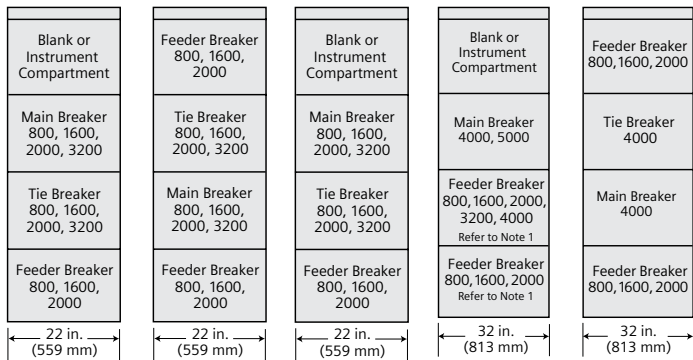
Tie Sections – Non-Fused Breakers



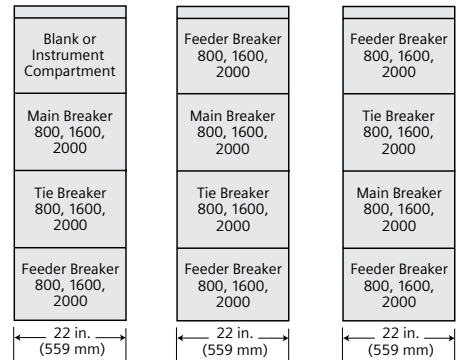
Tie Sections – Fused Breakers



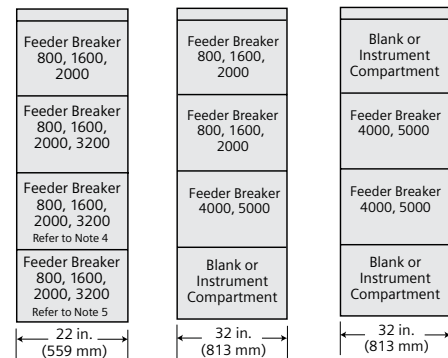
Main and Tie Sections – Non-Fused Breakers



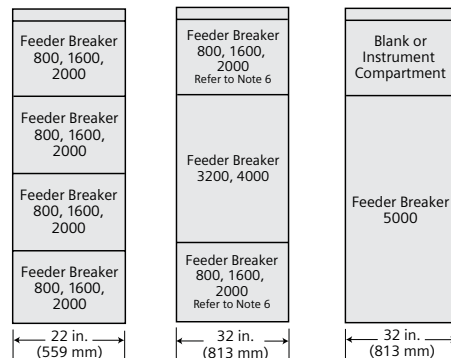
Main and Tie Sections – Fused Breakers



Feeder Sections – Non-Fused Breakers (see Note 7 on page 24)



Feeder Sections – Fused Breakers (see Note 7 on page 24)



Type WL Low Voltage Metal-Enclosed Switchgear

Shipping Weights and Dimensional Information

Siemens Type WL Low Voltage Switchgear can be configured in many ways by combining different section types. Up to five vertical sections plus a transition section can be shipped together as a unit. Maximum shipping split length for indoor structures is 110 in. (2794 mm). If all vertical sections are not to be shipped as a unit, specifications need to be provided that describe the limiting factors (e.g., low door or narrow hallway).

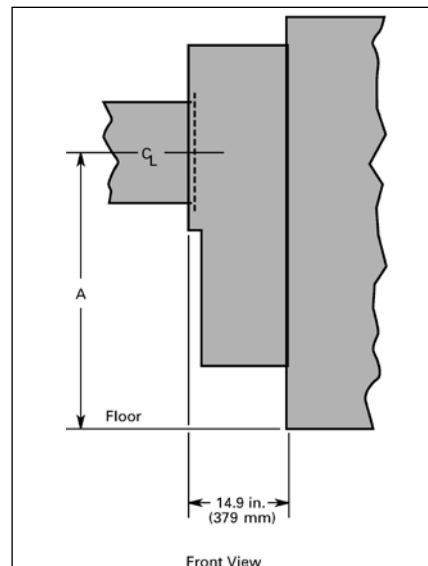
Normal indoor vertical sections are 96 in. (2438 mm) high and a minimum 60 in. (1524 mm) deep for non-fused breakers and 65 in. (1651 mm) deep for fused breakers. A top-mounted hoist, which is shipped as an accessory in a separate container, adds 6.2 in. (157 mm) for a total installed height of 102.2 in. (2596 mm).

The outdoor switchgear assembly contains the indoor assembly in an outdoor housing. The overall height is 112.8 in. (2865 mm) for non walk-in designs and 114 in. (2896 mm) for walk-in designs. The depth of a non walk-in outdoor

assembly with a 60 in. (1524 mm) internal structure is 82.3 in. (2090 mm) and the depth of a walk-in outdoor assembly with a 60 in. (1524 mm) internal structure is 110.7 in. (2812 mm). Maximum shipping split length for outdoor structures is 66 in. (1676 mm).

The major assembly sections include:

- Transition Sections — used as transition to liquid filled transformer or to outdoor dry type transformers.
- Auxiliary Sections — used as incoming bus duct or cable entrance when a main breaker is not used.
- Main Sections — used to contain main breaker and may house metering and feeder breakers.
- Feeder Sections — used to contain feeder breakers and other equipment such as instrumentation.
- Tie Sections — used to contain tie breaker and other equipment such as feeder breakers.



Transition Section For Liquid Filled and Outdoor Dry Type Transformers

	Dimension A in inches (mm)	Weight in lbs. (kg)
Indoor	55 (1397)	500 (227)
Outdoor	61 (1549)	550 (250)

Approximate Weight – Lbs.

Section Type	22" Indoor	22" Outdoor	32" Indoor	32" Outdoor	38" Indoor	38" Outdoor	48" Indoor	48" Outdoor
Auxiliary	1000 (450)	2000 (900)	1300 (585)	2500 (1125)	1800 (810)	3200 (1440)	N/A	N/A
Utility Metering	N/A	N/A	N/A	N/A	2100 (945)	3500 (1575)	2600 (1170)	4500 (2025)
Breaker	1400 (630)	2400 (1080)	2000 (900)	3300 (1485)	N/A	N/A	N/A	N/A

Weights shown in pounds and () kilograms.

Weights shown do not include weight of circuit breaker removeable element (but does include cradle).

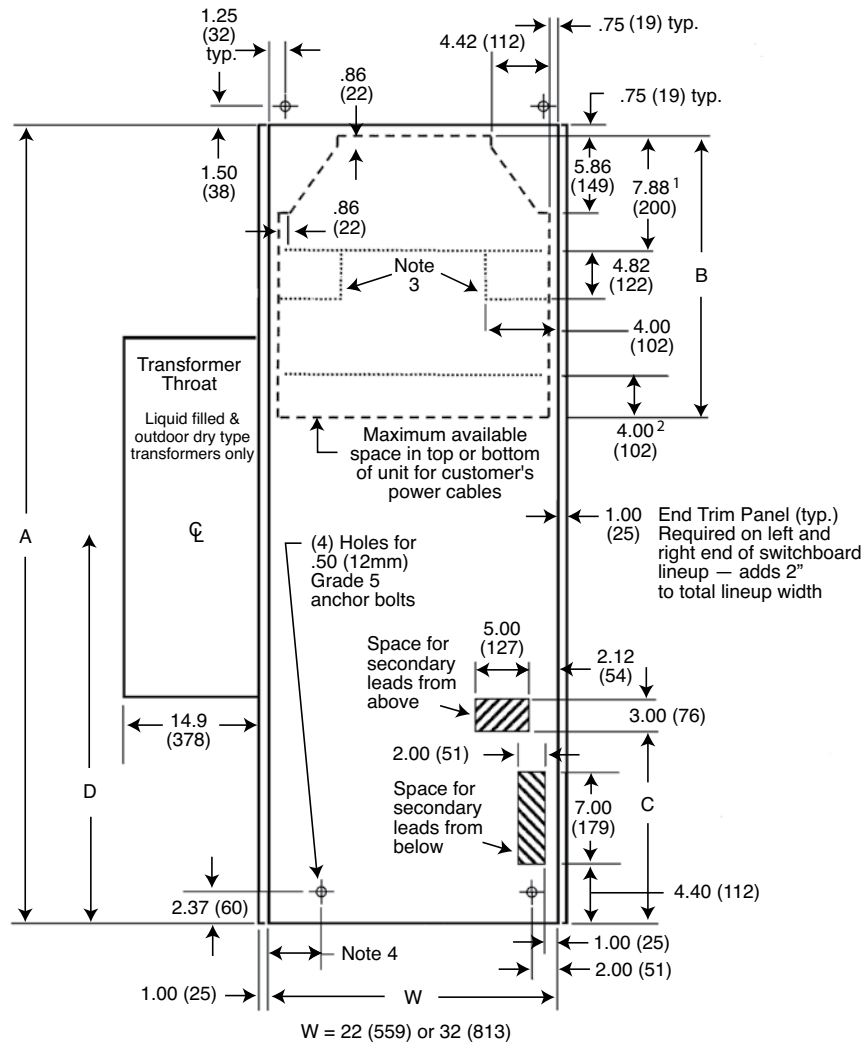
Add 400 lbs for hoist and track.

On outdoor switchgear, add 500 lbs for end walls (weight is for both ends). Refer to shipping documents for actual weights.

Type WL Low Voltage Metal-Enclosed Switchgear

Dimensional Information

Indoor Floor Plan and Cable Space Details



A Equipment Depth	Direction of Cables	B	C	D
60" Non-Fused with (N, S, H or L-Class Breakers) OR 65" Fused with (F-Class Breakers)	Below	21.50 (546) ^{1 2}	13.88 (353)	32.59 (828)
	Above	21.25 (540) ¹	18.88 (480)	37.59 (955)
70" Non-fused with (N, S, H or L-Class Breakers) OR 75" Fused with (F-Class Breakers)	Below	31.50 (800) ^{1 2}	13.88 (353)	32.59 (828)
	Above	31.25 (794) ¹	18.88 (480)	37.59 (955)
80" Non-fused with (N, S, H or L-Class Breakers)	Below	41.50 (1054) ^{1 2}	13.88 (353)	32.59 (828)
	Above	41.25 (1048) ¹		
80" Fused with (F-Class Breakers)	Below	36.50 (927) ^{1 2}	18.88 (480)	37.59 (955)
	Above	36.25 (921) ¹		

Note: Dimensions shown in inches and (mm).

¹ Reduce by 7.88" if upper neutral is present with cables above or if a lower neutral is present with cables below.

² Reduce by 4.00" if an 800-3200A breaker is located in the bottom compartment.

Reductions per notes 1 & 2 are additive. Example: cables below + lower neutral + 2000A breaker in bottom compartment = B-11.88.

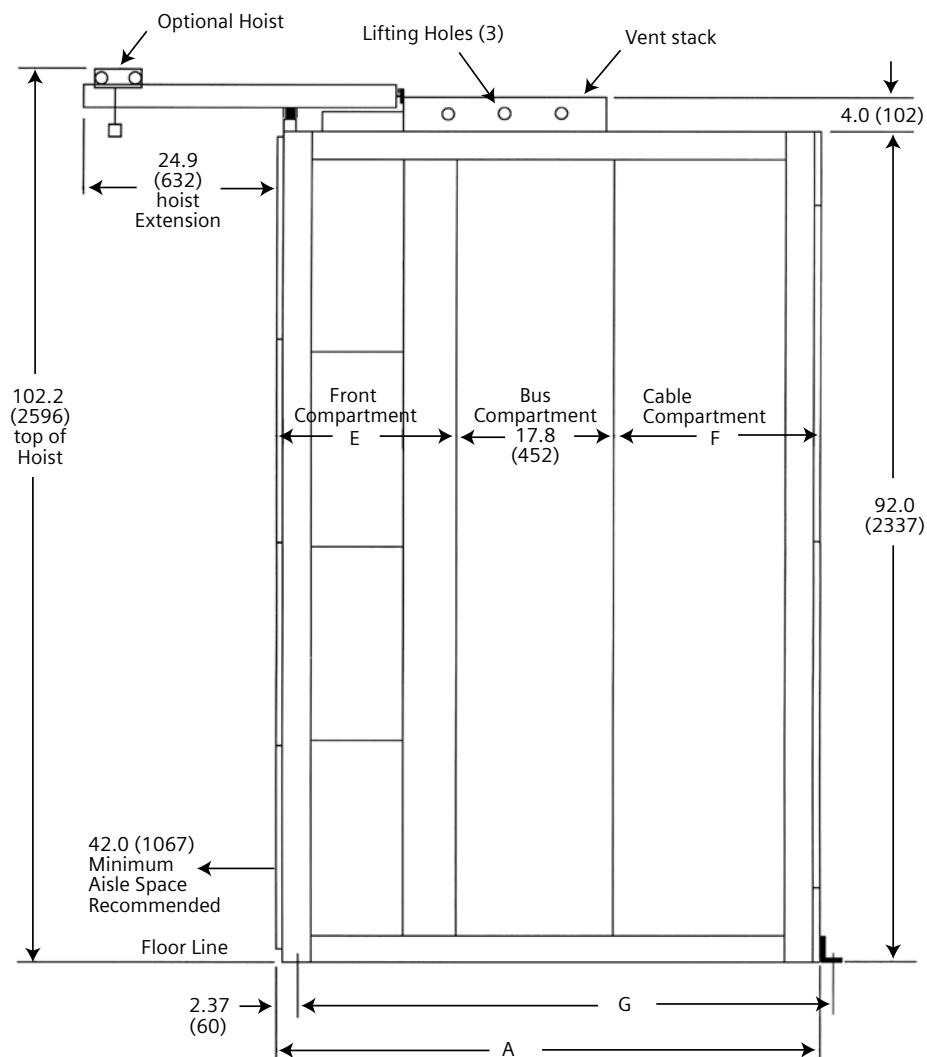
³ Reduce cable space by 4.00" x 4.82" if Neutral Riser is present. (Consult Factory).

⁴ 4.10 (104) if W=22; 4.60 (117) if W=32.

Type WL Low Voltage Metal-Enclosed Switchgear

Dimensional Information

Indoor Side View



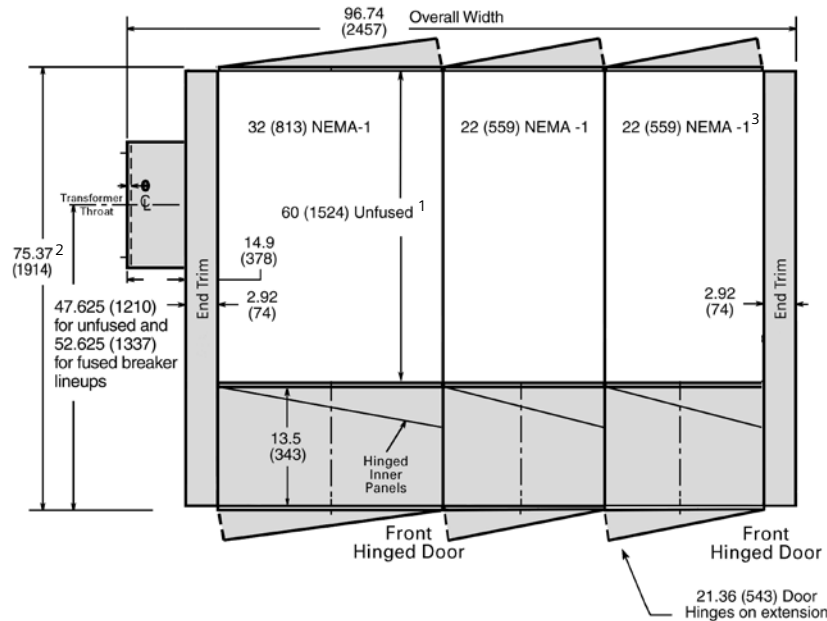
A	E	F	G
Equipment Depth	Breaker Compartment Depth	Rear Compartment Depth	Anchor Bolt Spacing
60 (1524) Non-fused breakers	19.8 (503)	22.4 (569)	59.13 (1502)
65 (1651) Fused breakers	24.8 (630)	22.4 (569)	64.13 (1629)
70 (1778) Non-fused breakers	19.8 (503)	32.4 (823)	69.13 (1756)
75 (1905) Fused breakers	24.8 (630)	32.4 (823)	74.13 (1883)
80 (2032) Non-fused breakers	19.8 (503)	42.4 (1077)	79.13 (2010)
80 (2032) Fused breakers	24.8 (630)	37.4 (950)	79.13 (2010)

Note: Dimensions shown in inches and (mm).

Type WL Low Voltage Metal-Enclosed Switchgear

Dimensional Information

Outdoor Non-Walk-in Floor Plan

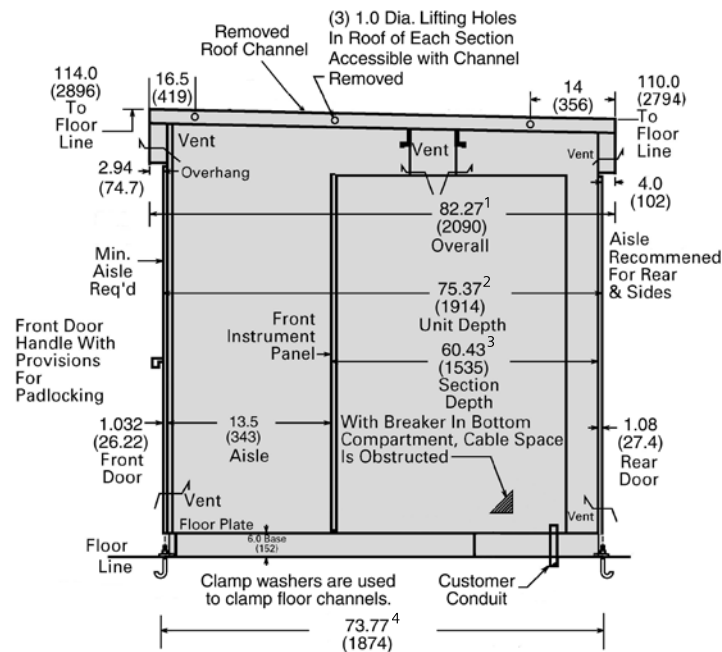


¹ 60" is representative for a 60" deep switchgear internal structure. For other internal structure depths (65 or 75) add extra depth to 60" that is shown.

² 75.37 is representative for a 60" deep internal structure. For other internal structure depths (65 or 75) add extra depth to 75.37 that is shown.

³ Refer to appropriate indoor plan view for available customer conduit information.

Outdoor Non Walk-in Side View



Dimensions shown in inches (mm)

¹ 82.27 dimension is based on 60" internal frame structure and if a deeper internal frame structure is used (65 or 75) the extra depth should be added to the 82.27 dimension.

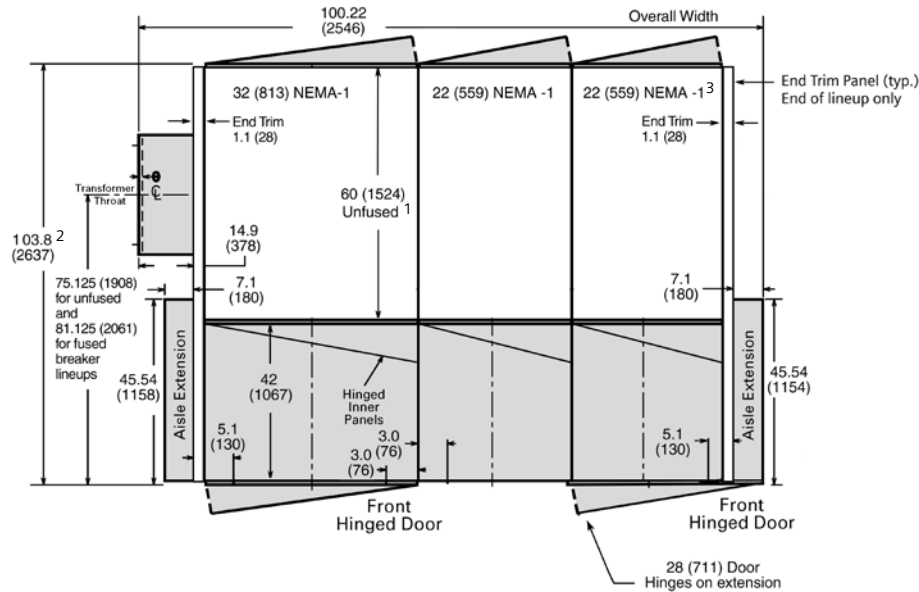
² 75.37 dimension is based on 60" internal frame structure and if a deeper internal frame structure is used (65 or 75) the extra depth should be added to the 75.37 dimension.

³ 60.43 dimension is based on 60" internal frame structure and if a deeper internal frame structure is used (65 or 75) the extra depth should be added to the 60.43 dimension.

⁴ 73.77 dimension is based on 60" internal frame structure and if a deeper internal frame structure is used (65 or 75) the extra depth should be added to the 73.77 dimension.

Dimensional Information

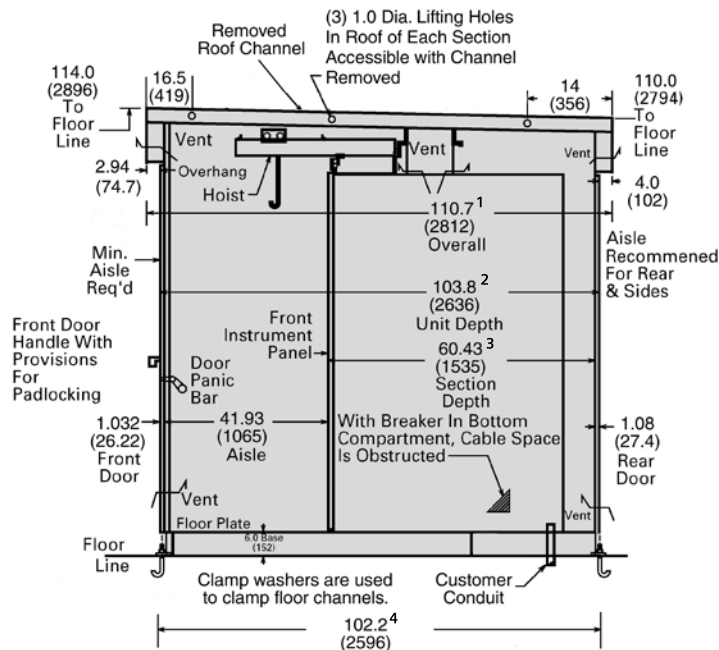
Outdoor Walk-in Floor Plan



- 1 60" is representative for a 60" deep switchgear internal structure. For other internal structure depths (65 or 75) add extra depth to 60" that is shown.
- 2 103.8" is representative for a 60" deep internal structure. For other internal structure depths (65 or 75) add extra depth to 103.8" that is shown.

³ Refer to appropriate indoor plan view for available customer conduit information.

Outdoor Walk-in Side View



Dimensions shown in inches (mm)

¹ 110.7 dimension is based on 60" internal frame structure and if a deeper internal frame structure is used (65 or 75) the extra depth should be added to the 110.7 dimension.

2 103.8 dimension is based on 60" internal frame structure and if a deeper internal frame structure is used (65 or 75) the extra depth should be added to the 103.8 dimension.

3 60.43 dimension is based on 60" internal frame structure and if a deeper internal frame structure is used (65 or 75) the extra depth should be added to the 60.43 dimension.

4 102.2 dimension is based on 60" internal frame structure and if a deeper internal frame structure is used (65 or 75) the extra depth should be added to the 102.2 dimension.

Type WL Low Voltage Metal-Enclosed Switchgear

VT, CPT, CT Data

Voltage Transformers – External Metering and Relaying

Ratio	Accuracy Class at 60 Hz			Volt-Amp Rating	Thermal Rating VA	Hertz
	Burden					
	W	X	Y			
600:120	0.6	1.2	1.2	100	150	50/60
480:120	0.6	1.2	1.2	100	150	50/60
288:120	0.6	1.2	1.2	100	150	50/60

Control Power Transformers – 115°C Rise

kVA	Phase	Primary Voltage	Secondary Voltage
3 5 10 ¹ 15 ¹	Single	240/480	120/240

Current Transformers for FSII WL Breaker Applications - External Metering and Relaying ²

Ratio	Accuracy at 60 Hz Metering Burden (ohms)					Class
	B-0.1	B-0.2	B-0.5	B-0.9	B-1.8	
100.5	1.2	—	—	—	—	C5
150.5	1.2	—	—	—	—	C7
200.5	1.2	—	—	—	—	C9
250.5	1.2	—	—	—	—	C12
300.5	0.6	0.6	—	—	—	C15
400.5	0.6	0.6	1.2	—	—	C20
500.5	0.6	0.6	1.2	—	—	C25
600.5	0.3	0.3	0.6	1.2	1.2	C21
800.5	0.3	0.3	0.6	0.6	1.2	C29
1000.5	0.3	0.3	0.6	0.6	1.2	C35
1200.5	0.3	0.3	0.3	0.6	0.6	C20
1500.5	0.3	0.3	0.3	0.3	0.6	C25
1600.5	0.3	0.3	0.3	0.3	0.6	C27
2000.5	0.3	0.3	0.3	0.3	0.3	C34
2500.5	0.3	0.3	0.3	0.3	0.3	C20
3000.5	0.3	0.3	0.3	0.3	0.3	C21
3200.5	0.3	0.3	0.3	0.3	0.3	C20

Current Transformers for FSIII WL Breaker Applications – External Metering and Relaying ²

Ratio	Accuracy at 60 Hz Metering Burden (ohms)					Class
	B-0.1	B-0.2	B-0.5	B-0.9	B-1.8	
2000.5	0.3	0.3	0.3	0.3	0.3	C20
2500.5	0.3	0.3	0.3	0.3	0.3	C20
3000.5	0.3	0.3	0.3	0.3	0.3	C20
3200.5	0.3	0.3	0.3	0.3	0.3	C20
4000.5	0.3	0.3	0.3	0.3	0.3	C20
5000.5	0.3	0.3	0.3	0.3	0.3	C20
6000.5	0.3	0.3	0.3	0.3	0.3	C20

¹ Requires complete compartment.

² Breaker compartment will accept 1 set of CT's each on top and bottom primary disconnects.

Type WL Low Voltage Metal-Enclosed Switchgear

Guide Form Specifications

SECTION [26 23 00] [16430] LOW VOLTAGE SWITCHGEAR, WL

PART 1 - GENERAL

1.1 SCOPE

- A. This section defines low voltage metal-enclosed switchgear assemblies with drawout circuit breaker elements for use in AC systems, rated 600 V or less.

1.2 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.
- B. Related Sections (where applicable) include the following:
1. Section [26 xx xx] [16xxx] - Medium Voltage Load Break Switches
 2. Section [26 12 16] [16271] - Dry Type Substation Transformers
 3. Section [26 12 19] [16271] - Liquid Type Substation Transformers
 4. Section [26 09 13] [16290] - Electrical Power Monitoring and Control
 5. Section [26 43 13] [16289] - Transient-Voltage Suppression for Low Voltage Electrical Power Circuits
 6. Section [26 05 73] [16055] - Coordination Study

1.3 SUBMITTALS

- A. Submit shop drawings and product data for approval and final documentation in the quantities listed according to the Conditions of the Contract. Customer name, customer location and customer order number shall identify all transmittals.
- B. Documents for Approval: General arrangement drawings showing dimensioned elevation, floor plan, side view and foundation details, one-line diagram showing major features, nameplate legends, schematic diagrams and bill of material.
- C. Final Documents: Record documentation to include those in 1.03.B and wiring diagrams, list of recommended spare parts, instruction and installation manuals [and certified test reports]
- D. Product Data: Include features, characteristics and ratings of individual circuit breakers and other components. Also, time-current characteristic curves for over current protective devices, including circuit-breaker trip devices and fusible devices.

1.4 RELATED STANDARDS

- A. Comply with requirements of latest revisions of applicable industry standards, specifically including the following:
1. ANSI/IEEE C37.20.1 – Metal-Enclosed Low Voltage Power Circuit Breaker Switchgear
 2. ANSI/IEEE C37.50 – Test Procedure for Low Voltage AC Power Circuit Breakers Used in Enclosures
 3. ANSI/IEEE C37.51 – Conformance Testing of Metal-Enclosed Low Voltage AC Power Circuit Breaker Switchgear Assemblies
 4. ANSI/IEEE C37.13 – Low Voltage AC Power Circuit Breakers Used in Enclosures
 5. ANSI C37.16 – Preferred Ratings, Related Requirements and Application for Low Voltage Power Circuit Breakers and AC Power Circuit Protectors
 6. ANSI/IEEE C37.17 - Trip Devices for AC and General Purpose DC Low Voltage Power Circuit Breakers
 7. UL 1558 – Metal-Enclosed Low Voltage Power Circuit Breaker Switchgear
 8. UL 1066 – Low Voltage AC and DC Power Circuit Breakers Used in Enclosures
 9. NEMA SG5 – Power Switchgear Assemblies
 10. NEMA SG3 – Low Voltage Power Circuit Breakers
- B. Manufacturer Seismic Qualification: The low voltage switchgear shall meet and be certified to seismic requirements specified in the [IBC 2009 International Building Code] [CBC 2010 California Building Code] [ASCE American Society of Civil Engineers 7-10].
1. The low voltage general purpose dry type transformer shall be compliant with IBC 2009 parameters:
 - a. Building Occupancy Category (as defined in Table 1.1 from ASCE 2010): [I] [II] [III] [IV]
 - b. Seismic Design Category: [A] [B] [C] [D] [E] [F]
 - c. Site Class: [A – Hard Rock] [B – Rock] [C – Very dense soil and soft rock] [D – Stiff soil profile] [E – Soft Soil Profile] [F – Soil vulnerable to potential failure or collapse under seismic loading] as defined in IBC 2006 Table 1613.5.2 Site Class Definitions.
- d. Ip – Importance Factor: [1.5 – Components must function after an earthquake for life safety purposes (Building Occupancy Code IV)] [1.25 – Buildings and structures that represent a substantial hazard to human life in the event of failure or that can cause substantial economic impact or mass disruption of day-to-day civilian life (Building Occupancy Code III)] [1.0 – Non-essential buildings. Function not life critical. (Building Occupancy Code I and II)]
- e. Ss – Mapped Spectral Accelerations for Short Periods at 0.2 seconds – 180%g
- f. Sds – 5% Damped Design Spectral Response Accelerations for Short Periods at 0.2 seconds – 1.2.g. z/h – Height factor ratio: [] Note: Ratio is a calculated value equal to the floor the gear is installed on divided by 12. A 6th floor installation is a 0.5 value. A basement or ground floor installation is a 0.0 value.
2. Equipment shall be designed to be located in a concrete and steel, moment-resisting frame building not exceeding 12 stories in height with a minimum story height of 10 feet.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Manufacturer of this equipment shall have a minimum of 5 years experience producing similar electrical equipment. The manufacturer of the switchgear assembly shall be the same manufacturer as the breakers. The manufacturer shall be ISO 9001 or 9002 certified.

1.6 DELIVERY, STORAGE AND HANDLING

- A. Deliver products in factory labeled packages. Shipping groups shall not exceed 15 ft. in length.
- B. Circuit breakers shall be shipped inside their respective cells in which they were factory acceptance tested.
- C. Store and handle in strict compliance with manufacturer's instructions and recommendations. Protect from potential damage from weather and construction operations. Store so condensation will not form on or in switchgear and if necessary, apply temporary heat where required to obtain suitable service conditions.

Type WL Low Voltage Metal-Enclosed Switchgear

Guide Form Specifications

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. The low voltage switchgear shall be manufactured by Siemens, Type-WL Low Voltage Switchgear or pre-approved equal. Approved manufacturers are as follows:
1. SIEMENS - Type-WL Low Voltage Switchgear

2.2 RATINGS

- A. System Configuration: Switchgear suitable for application in three-phase, [60 Hz] [50 Hz], [3 wire] [4 wire] [grounded-neutral] [3 wire ungrounded] [3 wire high-impedance grounded] system.
- B. Electrical Ratings:
1. Nominal System Voltage: [600 V] [480 V] [240 V] [208 V] [OTHER].
 2. Maximum Design Voltage: [635 volts with non-fused breakers] [600 volts with fused breakers]
 3. Short-Circuit Current: [100kA] [150kA] [200kA]
 4. Main-Bus Continuous Current: [1600] [2000] [3200] [4000] [5000] [6000] A.
 5. [Neutral Bus Continuous Current:] [50] [100] percent of main phase bus.

2.3 GENERAL REQUIREMENTS

- A. The switchgear shall be factory assembled and tested and comply with applicable industry standards. It shall be a coordinated design so that shipping groups are easily connected together at the site into a continuous line-up. Necessary connecting materials shall be furnished. All power circuit breakers and assemblies shall be produced by a single manufacturer.
- B. The switchgear assembly shall consist of one or more metal-enclosed sections in an [indoor NEMA 1 enclosure] [outdoor NEMA 3R non walk-in enclosure] [outdoor NEMA 3R walk-in enclosure].
1. End sections shall include provisions for main bus extension and installation of future vertical sections.
 2. The design shall incorporate preformed steel channels, angles and structural components bolted together and reinforced to form a rigid, self-supporting assembly.
 3. Fabricate enclosure with removable, [rear cover panels, secured by captive screws], [hinged rear doors with captive screws] [hinged rear doors with three-point latch and padlockable handle] to allow access to rear interior of switchgear.

- C. Front breaker doors and covers must be free of any ventilation openings.
- D. Horizontal barriers are to be provided to form individual circuit breaker or metering compartments. Circuit breaker compartments are to be barriered from the bus compartment through a primary disconnect assembly. Each circuit breaker or metering compartment shall be provided with a hinged front door secured with hand-operated [lockable] rotary latches.
- E. Circuit breaker compartments shall include stationary primary contact disconnects that shall be silver-plated copper at the connection points and of one-piece construction.
1. The upper set of disconnects shall bolt directly to the main bus and, for feeder circuit breakers, the lower set shall extend to the rear cable area and shall be insulated where they pass through the main bus compartment.
 2. Primary disconnects shall be sized for the maximum continuous current for the frame size of the circuit breaker which is located in the compartment.
 3. Interlocks shall be provided to prevent a circuit breaker element of the incorrect frame size or interrupting rating from being inserted into the compartment.
 4. Secondary control and communication connections, when required, shall be located in a separately accessed area that is accessible from the front of the switchgear without opening the breaker compartment door or exposing any power cables or bussing. The secondary control contacts shall be of the sliding contact design, silver plated and engage the drawout circuit breaker element in the "connected" and "test" positions.
- F. All control wiring within the assembly shall be continuous and shall terminate on each end at a suitable terminal block. Control wiring shall be 14-gauge minimum, stranded type SIS and shall be labeled at each end with sleeve-type wire markers.
1. Wire markers shall be machine imprinted with the wire name as indicated on the wiring diagrams.
 2. Terminals shall be insulated locking fork or ring tongue type except where connecting to components that do not accept these terminations. Control wiring for external connections shall be terminated in a separate front accessible compartment for ease of access.

- G. Finish: Steel parts shall be prepared for painting by a five-stage wash system consisting of an alkaline cleaner, fresh water rise, iron phosphate treatment, fresh water rise and non-chromate sealer. After cleaning and stabilization, the steel parts shall be coated with a thermosetting polyester powder applied with electrostatic equipment at a nominal 2 mils dry film thickness and then cured properly. The paint finish shall have a pencil hardness of 2H, a salt spray rating as defined in ASTM B-117-73 of 600 hours. Paint color shall be ANSI 61 light gray.
- H. Bus isolation barriers shall be arranged to isolate the buses on either side of each main and tie circuit breaker from each other.
- I. [Incoming line isolation barriers shall be arranged to isolate the incoming line connections from the main horizontal and vertical bus].
- J. Main bus shall connect vertical sections and shall be uniform capacity the entire length of assembly. Vertical and horizontal bus bar shall utilize a channel shape design to maximize short circuit withstand capability and minimize heat rise. The main horizontal bus shall be run in a vertical, edge-to-edge arrangement for high short circuit strength. Access to the rear cable termination area shall be possible without reaching over the main and vertical bus.
1. Bus shall be [98 % minimum conductivity copper silver-plated over entire length of the bus bar] [98 % conductivity copper tin-plated over entire length of the bus bar].
 2. Feeder Circuit-Breaker Load Terminals: Plated copper bus extensions equipped with pressure connectors for outgoing circuit conductors.
 3. Ground Bus shall be copper of 98 % minimum conductivity, minimum size 1/4 by 3 inches.
 4. Bus bracing shall be equal to the short circuit interrupting rating of the lowest rated circuit breaker applied in the assembly or 100kA minimum.
 5. Provide for future extensions from either end of main phase, neutral and ground bus by means of predrilled bolt holes and connecting links.
- K. Bus/Cable compartment barriers: Barriers shall be supplied to isolate the rear cable area from the main bus area.

Type WL Low Voltage Metal-Enclosed Switchgear

Guide Form Specifications

- L. [Insulated bus bar shall consist of bus bars enclosed in factory-applied, flame-retardant UL recognized insulation system. Bolted bus joints shall be insulated with secure joint covers that can easily be removed and reinstalled.]
 - M. [Low Voltage High-Resistance Grounding System] [Installed in the switchgear assembly] [Contained in separate NEMA 1 enclosure] for use on [480 V] [600 V] [wye] [delta] source, including the following components:
 - 1. Meter relay (MR)
 - 2. 3 Indicating lights: white for "Normal," red for "Ground Fault" green for "Pulsing"
 - 3. "Reset" push button
 - 4. "Test" push button
 - 5. On - Off switches for system and pulse
 - 6. Repeat cycle timer, set to produce approx. 30 current pulses per minute
 - 7. Neutral grounding resistor assembly
 - 8. Pulse resistor assembly
 - 9. Test resistor assembly
 - 10. Relay for repeat cycle timer
 - 11. Alarm relay, with extra interlocks for remote alarm (59X)
 - 12. Control power transformer
 - 13. Neutral ammeter & current transformer
 - 14. Optional Portable clamp on ammeter
 - 15. [3 Grounding transformers for generating a neutral CPT] (For Delta System without a system neutral)
 - N. Outdoor NEMA 3R Non Walk-In enclosure shall consist of indoor switchgear assemblies in weather-resistant steel housing. Enclosure shall be painted steel, integral structural-steel base frame with factory-applied asphalt undercoating. Each section shall be equipped with the following features:
 - 1. Structural design and anchorage adequate to resist loads imposed by 125-mph wind.
 - 2. Space heater operating at [full] [one-half of] rated voltage, sized to prevent condensation.
 - 3. Louvers equipped with insect/rodent screen and filter and arranged to permit air circulation while excluding exterior dust and rodents.
 - 4. Internal aisle of sufficient width to permit protective-device withdrawal.
 - 5. Load center for distribution of power to lights, receptacles and heaters.
 - 6. Incandescent lighting receptacles with three-way switch.
 - 7. Duplex receptacle with ground fault protection.
 - O. Outdoor NEMA 3R Walk-In enclosure shall consist of indoor switchgear assemblies in weather-resistant steel housing having an operating aisle space of approximately 42 inches. Enclosure shall be painted steel, integral structural-steel base frame with factory-applied asphalt undercoating. Each section shall be equipped with the following features:
 - 1. Structural design and anchorage adequate to resist loads imposed by 125-mph wind.
 - 2. Space heater operating at [full] [one-half of] rated voltage, sized to prevent condensation.
 - 3. Louvers equipped with insect/rodent screen and filter and arranged to permit air circulation while excluding exterior dust and rodents.
 - 4. Internal aisle of sufficient width to permit protective-device withdrawal, disassembly and servicing in aisle.
 - 5. Aisle access doors with outside padlocking provisions and interior panic latches.
 - 6. Adequate incandescent lighting receptacles with three-way switch at each access door.
 - 7. A duplex receptacle with ground fault protection, at aisle access door.
 - 8. A load center shall be supplied for distribution of power to lights, receptacles and heaters.
 - 9. Aisle ventilation louvers equipped with insect/rodent screen and filter and arranged to permit air circulation while excluding exterior dust and rodents.
 - P. Remote Racking
 - 1. A remote racking device shall be supplied to allow qualified personnel to rack Siemens Type WL breakers into Connect, Test and Disconnect positions from up to 30 feet away from the breaker and outside the arc flash hazard boundary.
 - 2. The remote racking device shall support utilization on any Frame Size 2 or Frame Size 3 WL breaker (including fuse carriage on fused Frame Size 3 WL breaker).
 - 3. The remote racking device shall be portable and weigh less than 30 pounds (excluding cables and remote control panel).
 - 4. The remote racking device shall have integral torque overload sensing to prevent damage to the breaker racking mechanism.
 - 5. The remote racking device shall allow breaker to be racked to any position (disconnect, test, connect) regardless of the starting position of the breaker and without the need for user input as to the starting position.
 - 6. The remote racking device shall support field retrofit on Type WL Low Voltage Switchgear.]
- ### 2.4 COMPONENTS
- A. Instrument Transformers: Comply with IEEE C57.13.
 - 1. Potential Transformers: Secondary-voltage rating of 120 V and NEMA accuracy class of 0.6 with burdens of W, X and Y.
 - 2. Current Transformers: Ratios as indicated; burden and accuracy class suitable for connected relays, meters and instruments.
 - B. Multifunction Digital-Metering Monitors shall be UL-listed or recognized, microprocessor-based unit suitable for three or four wire systems. Units shall be mounted in the instrument compartment door and as follows:
 - 1. Incoming monitoring or main breakers:
 - a. The main(s) shall be monitored.
 - 1.) The metering shall be accomplished with a Siemens Model [[PAC3100] [PAC3200] [PAC4200] [9340] [9360] [9510] [9510ADR/RTU] [9610]] digital meter.
 - 2. Feeder breakers:
 - a. Each feeder circuit breaker shall be monitored.
 - 1.) The metering shall be accomplished with a Siemens Model [PAC3100] [PAC3200] [PAC4200] digital meters on each feeder.
 - 2.) Each trip unit shall provide an LCD display for metering.
 - C. Provision for Future Devices: Equip future circuit breaker compartments with rails, mounting brackets, supports, necessary appurtenances and bus connections.
 - D. Control Power Supply: Control power transformer supplying 120-V control circuits through secondary disconnect devices are to be dry-type transformers with primary and secondary fuses.

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1. Transformers shall be mounted in auxiliary compartments.
 2. Multiple source with control power transfer. Two control power transformers located in separate compartments with necessary interlocking relays shall be provided.
 - a. Each transformer shall be connected to line side of associated main circuit breaker.
 - b. Secondary windings connected through a relay or relays to control bus to affect an automatic control power transfer scheme.
 3. [24] [48] [125] volt DC battery system
 - E. Mimic Bus: Continuous mimic bus applied to front of switchgear, arranged in single-line diagram format, shall indicate the arrangement of the circuit breakers in the power circuit.
- ### 2.5 CIRCUIT BREAKERS
- A. Circuit breakers shall comply with the requirements of IEEE/ANSI C37.13/16/17/50, UL1066, NEMA SG3. All breakers shall be three-pole, 100% rated type WL low voltage power breaker manufactured by Siemens Industry, or approved equal.
 1. Circuit breaker element shall have connected, test and disconnected position indicators, spring charged/ discharged indicators and circuit breaker open or closed and ready to close indicators all of which shall be visible to the operator with the compartment door closed. It shall be possible to rack the circuit breaker element from the connected to the disconnected position with the compartment door closed otherwise known as "through the door drawout".
 2. Provide interlocks to prevent racking the circuit breaker unless the breaker is open.
 3. Racking handle shall be integral to the breaker.
 - B. Ratings: Interrupting up to 100 kA at 480V without fuses. Short time current ratings for each circuit breaker shall be as indicated on the drawings or data tables. Circuit breakers shall be 600-volt class.
 - C. Operating Mechanism: Mechanically and electrically trip-free, stored-energy operating mechanism with the following features:
 1. Normal Closing Speed: independent of both control and operator
 2. Electrical operator, field installable with manual charging
 3. Operations counter
 - D. Each low voltage power circuit breaker shall be equipped with self-powered, microprocessor-based trip-device to sense overload and short circuit conditions. The device shall measure true RMS current. The tripping system shall consist of Rogowski coil sensors on each phase, a release mechanism and the following features:
 1. Field Installable and interchangeable so that any trip unit can be used with any frame size circuit breaker. Trip units can be upgraded for future expansion in functionality, such as communication.
 2. Functions: Long time, short time and extended instantaneous protection function shall be provided {EIP} to allow the breaker to be applied at the withstand rating of the breaker with minus 0% tolerance so that there is no instantaneous override whatsoever. This feature shall furthermore allow the circuit breaker to be applied up to the full instantaneous rating of the breaker on systems where the available fault current exceeds the breakers withstand rating. Each shall have an adjustable pick-up setting. In addition, long time and short time bands shall each have adjustable time delay. Short time function shall include a switchable I2t ramp and optionally I4t to improve co-ordination with fuses or inverse relays.
 3. A software program shall be made available free of charge to support system co-ordination studies. The software will allow time current curves to be generated for the chosen settings.
 4. Individual LED's shall indicate an overcurrent, short circuit or ground fault trip condition. The data shall be maintained for a minimum of 48 hours without the need for a separate battery.
 5. Time-current characteristics shall be field adjustable locally or optionally remotely via a bus system [ModBus] [Profibus] [Ethernet].
 6. Current Adjustability: Dial settings and rating plugs on trip units.
 7. Pickup Points: 10 Long Time Settings.
 8. Field Installable Ground-fault protection with at least three time-delay bands. Adjustable current pickup and an I2t ramp. Arrange to provide protection for [three-wire] [four-wire] service.
 9. Field installable zone selective interlocking: Connections will be made between main, tie and feeder circuit breakers to ensure that the circuit breaker closest to the fault trips for short time and ground fault conditions.
 10. Field Installable Communications shall be provided per schedule.
 11. A LCD display shall be available to simplify settings & viewing data locally.
 12. The option to remotely switch protection settings shall be provided whenever a generator is part of the power distribution system.
 13. Field installable configurable [analog], [digital] output relays shall be available to connect directly to the trip unit
 14. Waveform display option on LCD display ETU776 only
 15. Estimated contact wear shall be capable of being communicated remotely in addition to a local mechanical indication flag.
 - E. MOC {mechanism operated cell switch} operated by the circuit breaker operating mechanism
 - F. Terminal Block Connections, shall be front mounted and utilize [Screw Type Terminals], [Ring Tongue Terminals], [Tension Spring Terminals]
 - G. Padlocking Provisions: For installing at least three padlocks on each circuit breaker to prevent movement of the drawout mechanism.
 - H. Operating Handle: shall be built in complete with handle and integral to breaker. No external tools shall be required to rack the breaker.
 - I. Control Switch: One for each electrically operated circuit breaker.
 - J. Key Interlocks: Mountings and hardware are included where future installation of key-interlock devices is indicated.
 - K. Undervoltage Trip – field installable: [Instantaneous] [Adjustable time-delay.]
 - L. Shunt-Trip – field installable
 - M. Fused Circuit Breakers: Circuit breaker and fuse combinations must comply with requirements for circuit breakers and trip devices and include the following:

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1. Fuses: NEMA FU 1, Class L current limiting, sized to coordinate with and protect associated circuit breaker.
2. Blown-Fuse Trip Device: fused circuit breakers are to be equipped with blown fuse lockout devices to prevent closing the breaker if a fuse is blown or not present. Open-fuse status is indicated at the front of the circuit breaker or fuse drawout element.
- N. Indicating Lights: To indicate circuit breaker is open or closed, for electrically operated circuit breakers.
- O. Modular communication and relaying accessories are to be available for retrofitting by the clients chosen engineer. It shall not be necessary for the manufacturers personnel to retrofit accessories
- P. The following items must be capable of being changed in the field: main contacts, CT's, trip units, racking mechanism and all internal & external accessories
- Q. The main breaker shall have a Dynamic Arc Flash Sentry. The main breaker shall have a dual protective setting capability with graphic waveform display, similar to the Siemens WL breakers ETU776 trip unit. The main breaker will allow the installer to set two different trip curves into one breaker. One curve will be set for standard operating mode and the second curve, with instantaneous protection shall be set for arc flash mode. The switchboard shall be outfitted with a 24 VDC power supply, CubicleBus digital input module, annunciator panel with flashing light and a UPS power supply. The arc flash mode shall be actuated by a [Siguard motion sensor set to the arc flash boundary] [keyed switch] [selector switch]. [A light curtain shall be installed behind the gear to sense open doors on the back of the panel.]
- D. Spare Fuses: Six, of each type and rating of fuse used. Include spares for potential transformer fuses and control power fuses.
- E. Spare Indicating Lights: One of each type installed.
- F. Touchup Paint: One-half pint of paint matching enclosure finish.
- G. Test Cabinet: Wall mountable cabinet to hold necessary equipment for testing electrically operated breakers.
- H. Remote Breaker Racking
 1. Provide a portable remote breaker racking device system which consists of gear mounted brackets and a cable connected motor operator and control station.
 2. The 120v remote breaker racking device shall be attached to the factory mounted switchgear brackets (field retrofit capable) and secured by locking pins. No modification of the gear or circuit breakers shall be required for installation or operation.
 3. A single remote breaker racking device shall work on all circuit breaker frame sizes and draw out fuse frames.
 4. The remote breaker racking device shall have an on off power switch.
 5. The remote breaker racking device shall have torque control to prevent damage to the breaker racking mechanism
 6. An operator control station shall be on the end of 30 feet cable to assure the operator is able to locate them self safely.
 7. The operator control station shall have the capability of selecting the position desired for the breaker and shall include connected, test and disconnected positions.
 8. The operator control station shall contain Start-Stop buttons to initiate the breaker racking.
 9. The operator station shall contain indicator lamps showing connected, test and disconnect positions of the breaker as it is racked in or out.

2.6 ACCESSORIES

- A. Lifting yoke for circuit breakers.
- B. Portable test set for testing all functions of circuit breaker, solid-state trip devices without removal from switchgear.
- C. Circuit-Breaker Removal Apparatus:
 1. Overhead-circuit-breaker lifting device, track mounted at top front of switchgear, complete with hoist and lifting yokes. Portable, floor-supported, roller-base, elevating carriage arranged for moving circuit breakers in and out of compartments.

3.2 CONNECTIONS

- A. Tighten bus joints, electrical connectors and terminals according to manufacturer's published torque-tightening values. Install equipment-grounding conductors for switchgear with ground continuity to main electrical ground bus.

3.3 ADJUSTING AND CLEANING

- A. Set field-adjustable trip devices per coordination study.
- B. Clean exposed surfaces using manufacturer recommended materials and methods. Touch-up damaged coating and finishes using non-abrasive materials and methods recommended by manufacturer. Eliminate all visible evidence of repair.

3.4 WARRANTY

- A. Equipment manufacturer warrants that all goods supplied are free of non-conformities in workmanship and materials for one year from date of initial operation, but not more than eighteen months from date of shipment.

3.5 STARTUP SERVICES

- A. Engage a factory-authorized service representative to perform startup service.
- B. Train Owner's maintenance personnel on procedures and schedules for energizing and de-energizing, troubleshooting, servicing and maintaining equipment and schedules.
- C. Verify that switchgear is installed and connected according to the Contract Documents.
- D. Verify that electrical control wiring installation complies with manufacturer's submittal by means of point-to-point continuity testing. Verify that wiring installation complies with requirements in Division [26] [16] Sections.
- E. Complete installation and startup checks according to manufacturer's written instructions.

END OF SECTION

PART 3 - EXECUTION

3.1 EXAMINATION AND INSTALLATION

- A. Examine elements and surfaces to receive switchgear for compliance with installation tolerances and other conditions affecting performance. Proceed with installation only after unsatisfactory conditions have been corrected.
- B. Install and anchor switchgear in accordance with manufacturer's instructions.

Note: This guide does not purport to cover all details in equipment, or to provide for every possible contingency. Should further information be desired or should particular problems arise, which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the local Siemens sales office. The contents of this guide shall not become part of or modify any prior or existing agreement, commitment or relationship.

Siemens Industry, Inc.
5400 Triangle Parkway
Norcross, GA 30092

1-800-241-4453
info.us@siemens.com

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